



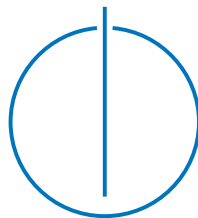
SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Information Systems

**Defining and evaluating characteristics of
Non-IID data in the context of Federated
Learning**

Ayşe Aybüke Ulusarslan





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Definieren und Bewerten von Non-IID-Merkmalen im Kontext des föderierten Lernens

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I confirm that this bachelor's thesis in information systems is my own work and I have documented all sources and material used.

Munich, 17.04.2023

Ayşe Aybüke Ulusarslan

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I would like to dedicate this work to the memory of Sabahat Ulusarslan and Güliz Cetinbakis. I know you would have been overjoyed to see me succeed, and I carry your memory with me as I complete this journey.

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Abstract

Federated learning is a promising machine learning framework where decentralized devices collaboratively train a shared model while keeping their data locally, reducing the need for centralized data storage and conceptually enhancing privacy and bandwidth performance. However, the absence of a globally aggregated training dataset presents significant challenges for general machine learning approaches, primarily due to data heterogeneity across clients, which invalidates the commonly employed IID assumption. With learning performance degrading under non-IID training data, considerable efforts have been made to develop improved learning techniques, yet no clear or universally accepted guideline exists for determining data dissimilarity. This thesis investigates various distance metrics and their correlation to trained model performance using systematically adjusted heterogeneous partitioned datasets. Notably, the entropy-based Kullback-Leibler divergence and Wasserstein distance metrics demonstrate a linear relationship to predictive accuracy, making them preferable choices for assessing data dissimilarity.

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1 Overview

1.1 Motivation

Federated Learning, first proposed in 2017 by McMahan et.al. [1], is a distributed machine learning framework with strong privacy guarantees which holds great promise for privacy-sensitive and safety-critical areas such as healthcare and autonomous driving. In contrast to the typical centralized machine learning setting, the training data as well as the training procedure itself is distributed among an ensemble of clients, who only have access to their own data. While the elimination of central data collection is at the heart of aforementioned privacy guarantees, and thus critical to the value proposition of Federated Learning, it imposes certain optimization challenges and necessitates a reevaluation of one of the most fundamental assumptions in Machine Learning: the IID assumption.

Although the assumption of all data points in the training set being independent and identically distributed (IID) rarely applies to any real-life scenario, the lack of it is less problematic in classical machine learning frameworks, where the data can be collected centrally to form a larger pool of training samples, and preprocessed such that it conforms to expected quality standards. In the Federated Learning setting, however, non-IIDness can be much more pronounced due to the massively distributed and isolated nature of training data. Also, clients samples are usually much smaller and can be tightly coupled to the behaviors and preferences unique to that individual Example. This may result in local data distributions that are vastly different from each other as well as their collective.

The only information shared by the clients is a gradient (descent) update, which a trusted central server receives and aggregates into a single global gradient update by computing a weighted average. In an ideal scenario where the clients data is IID, this weighted average would perfectly reflect the global and local distributions, but this is typically not the case for the Federated setting. Under strong statistical heterogeneity, the convergence rate as well as the generalization ability of the global model can suffer greatly. That being the case, devising methods for inferring the degree of heterogeneity without violating expected privacy standards, and algorithmically incorporating this information becomes key in broadening the appeal and applicability of Federated Learning.

For example, Hsieh et.al. [2] have demonstrated that the degree of distributional shift often determines the difficulty level of achieving good model accuracy in a Federated setting, which further emphasizes the relevance of this objective. Having reliable methods for quantifying distributional shift is crucial not only for improving the Federated Learning framework itself but also for providing practitioners with a key data point in terms of problem assessment. Thus, much of the recent work in this area has been devoted to Non-IID Learning and our goal here is to review and build upon that body of research to identify quantitative characteristics of non-IID data for the Federated Learning setting.

1.2 Contributions

Our contributions can be summarized as follows:

- To our knowledge, the body of research that addresses the question of whether Non-IID'ness in the Federated Learning context can be quantitatively defined is very limited, thus, our work represents one of the earlier efforts towards that objective.
- We provide a comprehensive analysis of the statistical heterogeneity problem both from a theoretical and practical perspective, as well as an overview of existing strategies in the broader context of Multi-View and Non-IID Learning.
- We construct a three-staged experiment where we cross-validate the relationship between statistical heterogeneity and ML model performance with different partitioning methods, distance metrics and model types/architectures that can also be used as a benchmarking framework for related future work.

2 Introduction

This chapter includes an overview of the notation and terminology that will be used throughout this study, as well as the background information and formal definitions necessary to accurately describe our problem statement. We also briefly touch upon our evaluation approach at the end of this chapter.

2.1 Terminology

- We will assume that the training data is distributed over K clients, whom we will denote as $\{P_i\}_{i=1}^K$. In probabilistic discussions, the same notation will be used to describe the (hypothetical) data-generating probability distribution at client i . The collective of clients as well as the global data-generating probability distribution will be denoted as P .
- The local datasets will be denoted as $D_1, \dots, D_K$, where $D := \cup_{i=1}^K D_i$, where (in a supervised setting) each dataset contains pairs of input vectors and labels: $D_k = \{(x_{k,i}, y_{k,i})\}_{i=1}^{n_k}$.
- We will further assume that $P_1, \dots, P_K$ belong to the same family of probability distributions with shared sample space (X, Y) where $X \subseteq \mathbb{R}^d$ is the feature space and $Y \subseteq \mathbb{R}$ the label space.
- The prediction model will be denoted as a parametric function $f(x; \theta) : X \rightarrow Y$ that maps any (legal) feature vector $x \in X$ to a value in the label space, where $\theta \in \Theta$ and Θ denotes the parameter space. This notation will occasionally be shortened as $f_\theta(x)$.
- We will denote the loss function as $\ell(y, f(x; \theta)) : Y \times X \times \Theta \rightarrow \mathbb{R}$, which can be any function that maps all points in the sample space $Z = X \times Y$ to a scalar value, and constitutes an adequate metric for describing the distance between predicted and actual labels (such as cross-categorical entropy or mean squared error). The loss function will occasionally be shortened as $\ell_\theta(x, y)$.

- We will denote the expected loss (or risk) at client P_i , under model θ , as $R_i(\theta) : R_i(\theta) = \mathbb{E}_{x,y \sim P_i}[\ell(y, f(x; \theta))] := \int \int \ell(y, f(x; \theta)) * P_i(x, y) d(x, y)$.
- The empirical loss at client P_i will be denoted as $\hat{r}_i(\theta) = \mathbb{E}_{(x,y) \in D_i}[\ell(y, f(x; \theta))] = \frac{1}{|D_i|} * \sum_{(x,y) \in D_i} \ell(y, f(x; \theta))$. Note that $\hat{r}_i(\theta)$ is an unbiased estimator of $R_i(\theta)$ under the assumption that $D_i \stackrel{\text{iid}}{\sim} P_i$.
- The true minimum risk (or expected loss) at client P_i will be denoted as $R_i^* \in \min_{\theta} R_i(\theta)$. The empirical minimum given dataset D_i will be denoted as $r_i^* \in \min_{\theta} \hat{r}_i(\theta)$.
- The true minimum of global risk will be denoted as $R^* \in \min_{\theta} \mathbb{E}_{x,y \sim P}[\ell(y, f(x; \theta))]$, and the empirical global risk minimum as $r^* \in \min_{\theta} \mathbb{E}_{(x,y) \in \bigcup_{i=1}^K D_i}[\ell(y, f(x; \theta))]$. Note that $D = \bigcup_{i=1}^K D_i$ isn't directly observed in the Federated setting.
- The update sent by client P_i (obtained through gradient descent) will be denoted as $\nabla_{\theta} r_i = \frac{1}{|D_i|} * \sum_{(x_i, y_i) \in D_i} \frac{\partial \ell(y_i, f(x_i, \theta))}{\partial \theta} := \Delta \theta_i$
- All random variables will be denoted with capital letters, while their values will be denoted with the corresponding small letter.

2.2 Federated Learning

Federated Learning, first proposed by McMahan et.al. [1], is a distributed machine learning framework, where an ensemble of participating devices collaboratively trains a global model. Under a centralized communication architecture, the collaborative training procedure is orchestrated by a trusted central server which is responsible for the maintenance of the global model state.

Federated Learning carries links to various fields such as Distributed Machine Learning, Edge Computing, Federated Database Systems, and Privacy-Preserving Machine Learning, but it is arguably unique in combining notions such as learning in a distributed environment, achieving time- and bandwidth-efficient communication by converging computational and data resources, integration of multiple autonomous data sources, and the usage of privacy-preserving mechanisms such as Differential Privacy [3] and encryption. It holds great promise for application domains such as healthcare and finance, where the privacy-sensitive and distributed nature of data makes classical Machine Learning frameworks less effective.

The existing body of research in Federated Learning can be broadly categorized into three distinct groups with different problem settings determined by data properties:

- **Vertical Federated Learning:** VFL refers to a setting where the data is vertically partitioned, i.e., where the sample space is shared with little intersection in feature space ($Z_i \neq Z_j, \mathcal{I}_i = \mathcal{I}_j \forall i, j \in [K], i \neq j$, where Z denotes the sample space and $\mathcal{I}$ the ID space, i.e., the set of identifiers for each node.). A typical real-world example for this setting is a collaboration between two commercial entities (e.g., a marketing company and an e-commerce company) who have data on the same set of subjects but with a disjoint sets of features/measurements.
- **Horizontal Federated Learning:** HFL refers to the setting where the feature space is shared, with little to no overlap between the sample spaces of the participants. ($Z_i = Z_j, \mathcal{I}_i \neq \mathcal{I}_j \forall i, j \in [K], i \neq j$). This corresponds to a learning scenario where different parties, each with their own set of data, collaborate to learn a single task. A typical example would be mobile keyboard prediction [4], where distributed mobile device data is used in a federated fashion to train a common neural network language model for word prediction.
- **Hybrid FL/Federated Transfer Learning:** Lastly, it is also quite common to have both disjoint ID spaces and a vertical data partitioning ($Z_i \neq Z_j, \mathcal{I}_i \neq \mathcal{I}_j \forall i, j \in [K], i \neq j$). In that case, the process is usually involves reducing the ID spaces such that $\mathcal{I}'_i = \mathcal{I}'_j \forall i, j \in [K], i \neq j$, and searching a shared representation that captures all features in this reduced ID space. Such a shared representation can be constructed by leveraging the principles of Transfer Learning/Inductive Transfer, which Liu et.al. [5] were first to adapt to the Federated setting in their paper from 2020.

Since different problem definitions necessitate different solution approaches, the Federated Learning framework has grown to encompass many different system architectures, model types and optimization technique. Some of the most prevalent aspects used for characterizing the different FL settings include communication design (Centralized vs. decentralized communication refers to whether the role of “trusted aggregator” is reserved for a single central entity or uniformly shared between the nodes), model type (Neural Networks, Decision Trees, . . .), and the privacy mechanism (e.g., Differential Privacy, Cryptographic Methods, etc.).

For the purposes of this study, we will assume a Horizontal Federated Learning setting since it best aligns with our focus on cross-device statistical heterogeneity and is the predominant choice in previous research related to our problem statement. We will further assume a centralized communication architecture (i.e., star topology).

2.2.1 The Algorithm

According to McMahan et.al. [1], the federated learning problem can be summarized as finding a global parameter configuration Θ such that the average empirical risk over all participating clients is minimized:

$$\min_{\theta} \hat{R}(\theta), \text{ where } \hat{R}(\theta) := \frac{1}{K} \sum_{i=1}^K \hat{r}_i(\theta) \quad (1)$$

and $\hat{r}_i(\theta) = \frac{1}{|D_i|} \sum_{(x_i, y_i) \in D_i} \ell(y_i, f(x_i; \theta))$ (empirical risk under the current parameter θ , given dataset D_i).

A slightly different formulation that takes potential differences in sample sizes into account would be:

$$\min_{\theta} \hat{R}(\theta), \text{ where } \hat{R}(\theta) := \sum_{i=1}^K w_i * \hat{r}_i(\theta) \text{ and } w_i = \frac{|D_i|}{\sum_{j=1}^K |D_j|} \quad (2)$$

Table 2.1: FedSGD: ServerProcess and ClientProcess Algorithms

Algorithm 1 ServerProcess(h, T, C_g)	Algorithm 2 ClientProcess(k, θ_t)
<pre> 0: $\theta_0 \leftarrow$ Initialize the model 0: for $t = 1, 2, \dots, T$ do 0: $K \leftarrow \max(C_g * N, 1)$ 0: $P^t \leftarrow$ Select K clients at random 0: for each client in P^t do 0: $(\Delta\theta_t^k, n_k) \leftarrow$ ClientProcess(k, θ_t) 0: end for 0: $n \leftarrow \sum_{k=1}^K n_k$ 0: $\Delta\theta_t \leftarrow \frac{1}{n} \sum_{k=1}^K n_k * \Delta\theta_t^k$ 0: $\theta_{t+1} \leftarrow \theta_t - h * \Delta\theta_t$ 0: end for </pre>	<pre> 0: $\theta_t^k \leftarrow \theta_t$ 0: $n_k \leftarrow D_k$ 0: for $(x_i, y_i) \in D_k$ do 0: $\Delta\theta_t^i \leftarrow \nabla_{\theta} l(y_i, f(x_i; \theta_t))$ 0: end for 0: $\Delta\theta_t^k \leftarrow \frac{1}{n_k} \sum_{i \in [n_k]} \Delta\theta_t^i$ 0: return $(\Delta\theta_t^k, n_k) = 0$ </pre>

The base algorithm FedSGD (or FedOPT) achieves this objective in an iterative manner over multiple communication rounds $t = 1, 2, \dots$: The server initializes and broadcasts the current model state to a randomly selected set of clients $P_1, \dots, P_k$. The clients then perform gradient descent on (a randomly selected, $C * |D_i|$ -sized batch of) their own data and send the average gradient back to the server. The server computes a weighted average of the received gradients $\{\nabla\theta_t^1, \dots, \nabla\theta_t^K\}$ and uses this average gradient (times the learning rate η) to update the global model as: $\theta_{t+1} \leftarrow \theta_t - \eta \cdot \nabla\theta_t^{Avg}$.

For simplicity, we will assume without loss of generality that the local gradient descent is non-stochastic ($C = 1$), and sample sizes are equal for all clients ($|D_i| = |D_j|$ for all $i, j \in [K]$ and $i \neq j$).

2.2.2 Federated Averaging

Although FedSGD serves as the baseline, in practice it is another federated optimization algorithm called Federated Averaging (FedAVG) that has established itself as the standard approach. It has been suggested by McMahan et.al. [1] as an iteration on FedSGD to improve generalization performance and convergence rates under increased statistical heterogeneity.

The main difference between FedSGD and FedAVG lies in the aggregation step: At each round of Federated Averaging, instead of sending the local gradient back to the server, the clients first apply it to the current model. This local gradient update step is repeated over E epochs (i.e., E steps are taken in the direction of steepest descent), and the issuing model state is sent back to the server for aggregation. So, in short, the order of aggregation vs. gradient update is reversed in Federated Averaging, and the object of aggregation is model updates instead of gradient updates.

Also, there is an additional (variance-reduction) step that precedes SGD where the clients' datasets are partitioned into smaller batches (minibatch-SGD). These so-called minibatches have a fixed size of B , which serves as a global hyperparameter. During each epoch $e \in E$, $\lceil \frac{n_k}{B} \rceil$ gradients are computed on the basis of each minibatch and applied to the current model: $\theta_{t+1}^k \leftarrow \theta_t^k - h \cdot \Delta\theta_t^b$.

Thus, the amount of computation performed by each client is proportional to the number of epochs (also a global hyperparameter) and inversely proportional to the size of the minibatches: $u_k = E \cdot \lceil \frac{n_k}{B} \rceil$. Note that Federated Averaging with the hyperparameter allocation $B = \infty$, $E = 1$ corresponds to FedSGD (as long as the entire dataset is included and only one step of gradient update is performed at each client, averaging locally updated models is equivalent to updating the same model with the average gradient). The amount of information that a single update contains remains unchanged as the gradients and the prediction model naturally have the same dimensions.

By allowing the clients to take multiple steps in the direction of the local gradients, the algorithm effectively shifts the computational load more towards the edge devices. Under bounded distributional shift and an appropriately chosen number of epochs E , the algorithm has been shown to converge to superior test accuracies in fewer communication rounds compared to FedSGD (McMahan et.al. [1]). More importantly for us, it has been shown to be more robust in the presence of highly unbalanced and non-IID data.

Table 2.2: FedAVG: ServerProcess and ClientProcess Algorithms

Algorithm 3 ServerProcess(h, T, C_g, B, E)	Algorithm 4 ClientProcess(k, h, θ_t, B, E)
<pre> 0: $\theta_0 \leftarrow$ Initialize the model 0: for $t = 1, 2, \dots, T$ do 0: $K \leftarrow \max(C_g * N, 1)$ 0: $P^t \leftarrow$ Select K clients at random 0: for each client in P^t do 0: $(\theta_{t+1}^k, n_k) \leftarrow$ ClientProcess(k, θ_t) 0: end for 0: $n \leftarrow \sum_{k=1}^K n_k$ 0: $\theta_{t+1} \leftarrow \frac{1}{n} \sum_{k=1}^K n_k * \theta_{t+1}^k$ 0: end for </pre>	<pre> 0: $\theta_t^k \leftarrow \theta_t$ 0: $n_k \leftarrow D_k$ 0: $\mathcal{B} \leftarrow$ split D_k into batches of size B 0: for $e \in E$ do 0: for $b \in \mathcal{B}, (x_i, y_i) \in b$ do 0: $\Delta \theta_t^i \leftarrow \nabla_{\theta} l(y_i, f(x_i; \theta_t))$ 0: end for 0: $\Delta \theta_t^b \leftarrow \frac{1}{B} \sum_{i \in [B]} \Delta \theta_t^i$ 0: $\theta_{t+1}^k \leftarrow \theta_t^k - h * \Delta \theta_t^b$ 0: end for 0: return $\theta_{t+1}^k, n_k = 0$ </pre>

Definition 1: (IID'ness) Two random variables X and Y are considered independent, when their mutual information is zero (i.e., when knowing the state of X does not change our belief about the state of Y and vice versa).

A set of random variables $(X_1, \dots, X_n)$, is formally considered independent, when the joint probability density function of $X_1, \dots, X_n$ equals the product of marginal probability distributions:

$$f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \prod_{i \in [n]} f_{X_i}(x_i).$$

Equivalently when the joint cumulative distribution function can be factorized as:

$$F_{X_1, \dots, X_n}(x_1, \dots, x_n) = \prod_{i \in [n]} F_{X_i}(x_i).$$

Further, the random variables $(X_1, \dots, X_n)$ are formally considered identically distributed, when $\forall i, j \in [n], i \neq j, x \in X : f_{X_i}(x) = f_{X_j}(x)$.

We say that set of random variables $(X_1, \dots, X_n)$ is IID (independent and identically distributed) when both conditions are met.

Definition 2: (The IID Learning Problem) The classical supervised machine learning problem can be boiled down to three components:

1. A set of different measurements $X = (X_1, \dots, X_d)$, modeled as a vector of random variables, that carry information on a certain target outcome Y , also modeled

as a random variable. The relationship between these measurements and the target outcome is determined by their joint probability density function $f_{X,Y}(x, y) : X \times Y \rightarrow [0, 1]$.

2. A prediction model $f(x; \theta) : X \rightarrow Y$ from a parametric family of functions $\{f_X(\cdot; \theta) | \theta \in \Theta\}$, which maps each set of measurements $x \in X$ to a specific outcome $\hat{y} \in Y$.
3. A loss function that measures the distance between the ground truth and predictions produced by $f_\theta(x) : \ell(y, f_\theta(x) := \hat{y}) : Y \times \hat{Y} \rightarrow \mathbb{R}$.

The objective is to find a prediction function f_θ that minimizes the expected distance between predicted and actual outcomes (expected loss or risk):

$$\min_{f_\theta} \mathbb{E}[\ell(y, f_\theta(x))].$$

Under the assumptions that X (input/features) and Y (outcome/label) are real-valued and continuous random variables with a known joint probability density function $f_{X,Y}(x, y)$, and that the first-order moment exists, the expected loss can be calculated as:

$$\mathbb{E}[\ell(y, f(x; \theta))] = \int_{x \in X} \int_{y \in Y} \ell(y, f(x; \theta)) f_{X,Y}(x, y) dx dy \equiv R(\theta),$$

where $f_{X,Y}(x, y)$ denotes the joint probability density of the input and outcome.

The joint probability density function $f_{X,Y}(x, y)$ underlying the sample space is usually unknown, and expected loss can't be measured directly by using the formula above. But it can be approximated on the basis of a sample population (training data) that is assumed to be drawn IID from the same data-generating probability distribution $f_{X,Y}$. This principle of choosing a prediction model by minimizing sample loss is also referred to as empirical risk minimization [6].

Under the aforementioned IID'ness assumption, the sample mean is an unbiased estimator of the true population mean, and empirical risk can simply be measured as the average loss over the training set $\{(x_i, y_i)\}_{i=1}^n$:

$$R_{\text{emp}}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(x_i; \theta)) \quad \text{and} \quad \mathbb{E}[R_{\text{emp}}(\theta)] =_{iid} R(\theta). \quad (2.1)$$

The new objective is then to find a model θ that minimizes this empirical quantity: $\theta^* \in \arg \min_{\theta} R_{\text{emp}}(\theta)$, which is essentially an optimization problem. Most machine learning problems are non-convex, so no closed-form solution for the ERM objective $\min_{\theta} R_{\text{emp}}(\theta)$ exists. But it is possible to find an approximate solution through first-order optimization algorithms such as the gradient descent. The selected model θ can

then be evaluated on the basis of test/generalization error (i.e., empirical loss at θ over a dataset not yet "seen" by the model).

Definition 3: (The IID Assumption of Machine Learning) Classical machine learning algorithms assume that all data points in the training set D_{train} as well as the test set D_{test} are drawn independently from the same underlying distribution $P_{X,Y}$. This assumption can be broken down into two separate components, on which the soundness of the previously mentioned Empirical Risk Minimization principle depends:

1. *In-population IID'ness*: The datapoints within the training set are assumed to be i.i.d. sampled from some fixed but unknown distribution. Only under this assumption can sample statistics used in the ERM process be regarded as consistent estimators of their population counterparts:

$$\mathbb{E}[\theta_{\text{train}}^*] \in \arg \min_{\theta} \mathbb{E}_{(x,y) \sim P_{X,Y}} [\ell_{\theta}(x, y)] \quad (2.2)$$

Strong statistical heterogeneity (or "noisiness") of training data is often associated with optimization challenges.

2. *Between-population IID'ness*: Only if the training set is sufficiently representative of the underlying data-generating distribution, from which the target populations are also drawn, can the ensuing model consistently achieve satisfactory performance levels on previously unseen datasets. The ability of a machine learning model to adapt to such "new" datasets is often referred to as out-of-sample generalization. Perfect generalization in the context of Empirical Risk Minimization can be expressed as an equality between the expected values of local optima w.r.t. empirical loss: $\mathbb{E}[\theta_{\text{train}}^* - \theta_{\text{test}}^*] = 0$.

For the purposes of this study, we will focus on the second type of (Non-) IID'ness, where the training and target populations have different distributions, and assume the noise training data to be negligible.

Definition 4: (Violations of the IID Assumption)

Identicalness: As previously discussed, the identicalness assumption can be expressed as all data points in the training set having been produced by the same joint probability density function $p(X, Y)$ of input features and labels. Following the definition of conditional probability (or the chain rule), the joint probability density function can be factorized as $p(X, Y) = p(Y|X) * p(X)$ or $p(X, Y) = p(X|Y) * p(Y)$. With the help of this factorization, we can identify four distinct sources of Non-IID'ness:

1. $p_i(Y|X) = p_j(Y|X)$, $p_i(X) \neq p_j(X)$: The supervisory mechanism is shared; marginal distribution of the input features is different. This type of Non-IID'ness is often referred to as feature distribution skew or covariate shift. [30]

2. $p_i(X) = p_j(X)$, $p_i(Y|X) \neq p_j(Y|X)$: Marginal distribution of the input features is shared; distribution of labels under a fixed input feature allocation is different. This is often referred to as concept shift.
3. $p_i(X|Y) = p_j(X|Y)$, $p_i(Y) \neq p_j(Y)$: The a posteriori distribution of features within the same fixed class label is shared, the marginal distribution of labels is different. Non-IID'ness caused by an imbalance in label distributions is often referred to as label distribution skew or prior probability shift.
4. $p_i(Y) = p_j(Y)$, $p_i(X|Y) \neq p_j(X|Y)$: The marginal distribution of labels is shared; the a-posteriori distribution of features within a given class label different. Non-IID'ness caused by an imbalance in the posterior distribution of features is often referred to as concept drift.

Our focus here will lie on heterogeneity caused by differences in the marginal distribution $p(y)$ of labels, because this is the one that causes the most severe forms of model divergence, and is the easiest one to measure and communicate.

Independence: A violation of independence between two random variables X and Y can be formally expressed as: $p(Y|X) \neq p(Y)$ or $p(X, Y) \neq p(X) * p(Y)$. In the Machine Learning context, such a violation could be caused by the presence of confounding variables (often called a "spurious relationship/correlation"), sampling bias, temporal autocorrelation, or multicollinearity. In the Federated Learning setting, sampling bias could be an issue if there is significant correlation between client availability and local distributions [1].

Since this is a topic that has more to do with the subtleties of accurate statistical modeling and model interpretation in the broader sense, any discussions regarding the independence of data points will be left out. We will instead focus on the identicalness aspect, which has more relevance for the Federated Learning context due to its direct and measurable effects on model accuracy and convergence.

2.3 The Non-IID'ness of Federated Learning

Countless iterations on the base Federated Learning framework have been proposed since its conception in 2014, but most of them constitute variations of the Federated stochastic gradient descent (FedSGD) algorithm, and the core principle of local training is never abandoned. Thus, what is actually being computed is a collection of local objectives:

$$\{\theta_i^* = \underset{\theta}{\operatorname{argmin}} \mathbb{E}_{(x_i, y_i) \in D_i} [\ell(y_i, f(x_i; \theta))] : i \in [K]\} \quad (2.3)$$

In the ideal IID case, where the underlying distributions of local data are identical ($\forall i, j \in [K], i \neq j : P_i(x, y) = P_j(x, y)$), the local optima are identical to each other ($\forall i, j \in [K], i \neq j : \theta_i^* = \theta_j^* = \theta^*$) as well as to the global optimum obtained by averaging. We could use the Weak Law of Large numbers to articulate a slightly relaxed version of the IID case:

$$\forall \varepsilon > 0 : \lim_{n \rightarrow \infty} \mathbb{P}(|\bar{\theta}_n - \theta^*| \geq \varepsilon) = 0, \text{ where } \bar{\theta}_n := \frac{1}{n} \sum_{i=1}^n \theta_i^* \quad (2.4)$$

(The average of local minima converges in probability to the true minimum θ^*). The effect is that the trained model generalizes arbitrarily well to local distributions. In the non-IID case, there is high variability within the set of local minima $\{\theta_i^*, i \in K\}$, and θ_K isn't guaranteed to converge to the true objective:

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \frac{1}{K} \sum_{i=1}^K \mathbb{E}_{(x_i, y_i) \in D_i} [\ell(y_i, f(x_i; \theta))] \quad (2.5)$$

We can approximate the mean divergence from θ^* as the average optimality gap [8] over participating clients:

$$\frac{1}{K} \sum_{i=1}^K \hat{r}_i(\bar{\theta}_K) - \hat{r}_i(\theta^*) \quad (2.6)$$

(Since θ^* isn't directly observed and the underlying global distribution is unknown in a realistic scenario, this quantity isn't directly measurable either, but it provides a useful construct for analyzing the convergence behavior of Federated Learning in presence of statistically heterogeneous data.) As a consequence, both model accuracy and convergence rates deteriorate with increasing statistical heterogeneity.

2.3.1 Quantifying Non-IID'ness

The degree of statistical heterogeneity across the nodes is a crucial aspect of any distributed learning problem, and especially so for federated since the convergence of local training has been shown to be directly determined by this quantity [7]. In many cases, the convergence rates can be improved through heterogeneity-informed hyperparameter tuning, for example by choosing a smaller learning rate when strong heterogeneity is present. So, it remains an important objective to formulate a canonical measure for non-IID'ness that can be extracted without violating the privacy guarantees and isn't restricted to certain use cases.

Since the model states are synchronized at the beginning of each communication round and the starting point for SGD is the same for all clients, the diversity within the set of gradients sent by the clients at a given round can be expected to represent

the non-identicalness of underlying distributions. This is especially easy to visualize if we assume that the clients' data are drawn from the same parametric family of distributions, such that the location of stationary points are representative of the underlying distributions. Many of the existing non-IID'ness measures are based on this observation and have been articulated in the context of measuring the relationship between the convergence behavior of distributed learning algorithms and statistical heterogeneity.

One example of this is the work of Khaled et.al. [8], [9], who have developed a comprehensive theory of convergence of local Stochastic Gradient Descent under identical and heterogeneous data. They formalize the dissimilarity of the distributions P_i and P_j via the dissimilarity of the local (loss) gradients:

$$\forall i, j \in [K], \theta \in \Theta : \|\nabla_{\theta} \hat{r}_i(\theta) - \nabla_{\theta} \hat{r}_j(\theta)\| \gg 1 \quad (2.7)$$

The quantify non-IID'ness as the average local gradient norm, measured at the local optimum:

$$\sigma^2 \triangleq \frac{1}{K} * \sum_{i=1}^K \|\nabla_{\theta} \hat{r}_i(\theta^*)\| \quad (2.8)$$

Under the assumption that both the global and local loss functions are convex, the derivative of the loss function w.r.t. θ , measured at the global optimum θ^* must equal zero under perfect IID'ness. With increasing sample size $K \rightarrow \infty$, $\frac{1}{K} \sum_{i=1}^K \hat{r}_i(\theta^*)$ will tend towards zero. Thus, it can be expected that this quantity will reflect the degree of divergence from the IID scenario.

Another gradient-based dissimilarity measure has been formulated by Haddapour et.al. in [10] where they followed a similar objective. They define a Weighted Gradient Diversity to “quantify heterogeneity among local objectives to establish conditions on convergence in non-i.i.d. setting”.

This definition of data heterogeneity builds on the notion of Gradient Diversity, first proposed by Yin et.al. [11]. The notion of Gradient Diversity is based on the observation that the inner product between two vectors gets smaller with increasing dissimilarity of the vectors. (In the extreme case where the vectors are orthogonal to each other, their inner product will yield zero.) Thus, they define gradient diversity as:

$$\Delta_S(\theta) = \frac{\sum_{i=1}^K \|\nabla_{\theta} \hat{r}_i(\theta)\|_2^2}{\sum_{i=1}^K \|\nabla_{\theta} \hat{r}_i(\theta)\|_2^2 + \sum_{i,j \in [K], i \neq j} \langle \nabla_{\theta} \hat{r}_i(\theta), \nabla_{\theta} \hat{r}_j(\theta) \rangle} = \frac{\sum_{i=1}^K \|\nabla_{\theta} \hat{r}_i(\theta)\|_2^2}{\sum_{i=1}^K \|\nabla_{\theta} \hat{r}_i(\theta)\|_2^2} \quad (2.9)$$

where S denotes the sample $S = \{(x_1, y_1), (x_2, y_2), \dots\}$ and θ can be any value from the parameter space Θ . Note that:

$$\Delta_S(\theta) = \begin{cases} 1, & \text{when } \nabla_{\theta} \hat{r}_i(\theta) \perp \nabla_{\theta} \hat{r}_j(\theta) \forall i, j \in [K] \\ 0, & \text{when } \nabla_{\theta} \hat{r}_i(\theta) = \nabla_{\theta} \hat{r}_j(\theta) \forall i, j \in [K] \end{cases} \quad (2.10)$$

Haddapour et al.[10] go one step further in adapting this definition of gradient diversity to the Federated Learning setting by incorporating the local dataset sizes:

$$\Delta_S(\theta, w) := \frac{\sum_{i=1}^K w_i * \|\nabla_{\theta} \hat{r}_i(\theta)\|_2^2}{\|\sum_{i=1}^K w_i * \nabla_{\theta} \hat{r}_i(\theta)\|_2^2}, \text{ where } w_i = \frac{|D_i|}{\sum_{i=1}^K |D_i|} \quad (2.11)$$

In a further study by Wang et al. [14], non-IID'ness was quantified as the difference between the global and local gradients, under the assumption of δ -boundedness:

$$\|\nabla_{\theta} \hat{r}_i(\theta) - \nabla_{\theta} \hat{r}(\theta)\| < \delta_i, \text{ where } \hat{r}(\theta) := \frac{1}{K} * \sum_{i=1}^K \hat{r}_i(\theta) \text{ and } \delta = \sum_{i=1}^K w_i * \delta_i \quad (2.12)$$

Many of the existing studies that examine the relationship between statistical heterogeneity and convergence properties of Federated SGD use very similar definitions to express the variance among local objectives, which is assumed to have an upper bound. One such example can be found in a study by Yu et al. [16], where an assumption of "bounded variances" is made:

$$\exists \delta \text{ s.t. : } \frac{1}{K} * \sum_{i=1}^K \|\nabla_{\theta} \hat{r}_i(\theta) - \nabla_{\theta} \hat{r}(\theta)\|_2^2 < \delta, \forall \theta \in \Theta \quad (2.13)$$

Finally, we have a more lightweight definition of a "degree of heterogeneity" in a 2020 study by Li et al. [12]:

$$\Gamma := \hat{r}(\theta^*) - \sum_{i=1}^K w_i * r_i(\theta_i^*), \text{ where } \theta^* = \arg \min_{\theta} \hat{r}(\theta) \text{ and } \theta_i^* = \arg \min_{\theta} \hat{r}_i(\theta) \quad (2.14)$$

Although this quantity - the difference between the minimum global loss and the weighted average of minimal local losses - arguably reflects the non-identicalness of the underlying distributions, we find it less suitable as a universal measure due to its sensitivity to changes in the data and other factors like the choice of the loss function. Another problem with this definition is that the global and local optima can't be known a priori (and can't be inferred either if the local optimizations won't converge as a consequence of strong statistical heterogeneity), which brings us to the next discussion.

2.3.2 Desirable properties in a heterogeneity measure

Although trivial, we find it necessary to discuss some of the properties that a heterogeneity measure must satisfy before we move on to introducing our approach.

A mathematically sound, robust and viable non-IID'ness measure must...

- ...only make use of empirical / measurable quantities (e.g., any direct references to the true mean $\mathbb{E}_{(x,y) \sim P}$ should be avoided).
- ...only make use of information that is available to the central server in a Federated Learning setting (e.g., direct use of local data as well as any distributional information should be avoided).
- ...be demonstrably insensitive to small changes in the data (potentially measurable through Influence or Differential Privacy).
- ...allow comparison across different data types, machine learning schemes, and architectures.
- ...map to a real-valued finite interval.
- ...demonstrably converge to the lower bound for identical or near-identical datasets.
- ...conform to general properties of (statistical) distance measures such as symmetry, non-negativity (and optionally an adherence to the triangle inequality).

3 Related Work

This chapter is devoted to previous studies that have informed ours and is split into three parts: First we link the Non-IID Federated Learning problem to broader Machine Learning paradigms such as Transfer Learning and Domain Generalization and review related work in those domains. We return to the Federated Learning context in the second part in order to provide an overview of existing strategies in dealing with Non-IID'ness. Lastly, we review studies with similar problem formulations to ours.

Although our main interest here is the aspect of Non-IID'ness itself, the ultimate objective that motivated our object of study is the ongoing search for alternative Federated Learning paradigms that are robust in the presence of strong statistical heterogeneity. In that light, we will first give an overview of existing Machine Learning paradigms that incorporate and optimize for the Non-IID'ness aspect. This brief detour from our main subject will redeem itself by providing context for the consequent discussion of existing Non-IID Federated Learning approaches. The ML paradigms featured in this section are all to some degree related to the notion of transferability of learned representations between related but distinct domains.

3.1 Multi-task and transfer learning

Multi-task learning, first proposed in 1997 by Caruana et al. [13], refers to the scenario where multiple related tasks are learned simultaneously. It is built upon the idea that similarities and differences among related learning tasks can be exploited to achieve superior generalization performance compared to training multiple individual models. It can be described -vaguely, but intuitively - as an adaptive regularization mechanism that imposes a (set of) secondary performance constraint(s) on the objective function instead of a penalty term.

Multi-task learning makes the assumption that there is a common representation that captures the behavior of all subdomains up to a certain level of accuracy. The “vanilla” objective of MTL can be expressed as finding an optimal shared representation through risk minimization:

$$\min_{\theta} \sum_{i=1}^K r_i(\theta) := \mathbb{E}_{(x,y) \in T_i} [\ell(y, f_x; \theta)] \quad (3.1)$$

, where T_i denotes the i^{th} task which can be seen as a generalization of the underlying data-generating probability distributions $\{p_i(x), p_i(y|x)\}$. It is quite common to include a weighting parameter that controls task priority:

$$\min_{\theta} \sum_{i=1}^K w_i * r_i(\theta). \quad (3.2)$$

The more general multi-task learning problem is one where the parameter space Θ is divided into a set of shared and model-specific parameters $\Theta = \Theta_{sh} \cup \{\Theta_i\}_{i=1}^K$, where Θ_{sh} represents the structure common to all learning tasks and $\{\Theta_i\}_{i=1}^K$ represents those that are task-specific. The objective function can then be written as:

$$\min_{\Theta_{sh}, \Theta_1, \dots, \Theta_K} \sum_{i=1}^K \mathbb{E}_{(x,y) \sim D_i} [\ell(y, f_x; \{\Theta_{sh}, \Theta_i\})] := \sum_{i=1}^K \hat{R}_i(\Theta_{sh}, \Theta_i) \quad (3.3)$$

The multi-task learning objective differs from Federated Learning in that there are multiple target models instead of one global model which is expected to perform well across deployment settings related to multiple tasks. Still, a multi-task modeling approach can be used to describe the non-IID'ness inherent in most federated learning tasks.

Transfer learning is related to multi-task learning and refers to a scenario where “knowledge transfer” from a source domain/task T_{src} to a target domain T_{tgt} is sought. A key assumption in transfer learning is that data from the target domain T_{tgt} isn't available during training, to the effect that knowledge transfer can only happen sequentially. A perfect use case for transfer learning would be one where data collection directly in the target domain is restricted, but one where we have access to high-quality auxiliary information collected from the same population.

3.2 Non-IID Federated Learning

There has been a growing amount of research in addressing the statistical heterogeneity challenge in Federated Learning [14]–[16]. The first and larger camp of studies in this body of research establish Federated Averaging (FedAvg) as the baseline and attempt to optimize its convergence behavior or overall performance in the presence of strong statistical heterogeneity by following various strategies such as exploiting the relationship between systems and statistical heterogeneity (FedProx) or normalizing the

local gradients before aggregation (FedNova). On the other hand, there are studies that reframe the problem formulation in a way that explicitly models or optimizes for a heterogeneous environment by drawing insights from other domains, like distributionally robust optimization or transfer learning, that per definition operate in a heterogeneous environment.

3.3 Degrees of Non-IID'ness

There are many studies that focus on quantifying the effects of statistical heterogeneity on convergence rates and model performance in a Federated setting, where definitions for the degree of Non-IID'ness can be found. Some of these definitions, mostly based on some notion of gradient (dis-) similarity, have been discussed in 2.

Although gradient dissimilarity as an indicator of statistical heterogeneity between clients' datasets is perfectly adequate for the purpose of analyzing convergence guarantees, it may fall short of capturing the entire nature of Non-IID'ness present in a given Federated Learning scenario. With the ever-increasing computational power available to individuals, and a growing awareness towards privacy, Federated Learning can be expected to extend its reach to new and emerging domains. Thus, it is ever more important to invest effort in developing the language necessary for describing the different types and degrees of heterogeneity that may be present in any given Federated Learning scenario. We will devote this section to reviewing previous efforts along that direction.

3.3.1 Non-IID Index (NI)

The study by He et.al. [17] is among the few with a clear (although not primary) emphasis in measuring Non-IID'ness itself. Their primary focus is on the Non-IID Image Classification task, which they define as learning a feature extractor g_ϕ and a classifier f_θ (the model is broken down into a component that extracts a higher-dimensional representation of the data and a component that maps this derived representation to a value in the label space: $g_\phi \circ f_\theta : X \rightarrow Y$) under concept drift: $p_{train}(X|Y) \neq p_{test}(X|Y)$.

They then go on to define the degree of shift between the training and test sets for a given label C as:

$$NI(C) = \left\| \frac{\mathbb{E}[g_\phi(X_{train}^C) - g_\phi(X_{test}^C)]}{\text{Var}[g_\phi(X_{train}^C \cup X_{test}^C)]} \right\|_2 \quad (3.4)$$

which simply corresponds to the normalized distance between the class-conditional feature distributions of the training and test sets. Experimental results presented in the same study have shown strong correlation between the NI-Index and loss of

classification accuracy, which can be regarded as a practical confirmation of this metric's validity.

However, its applicability to the Federated setting is rather limited because, even if we assume an image classification scenario with model (instead of gradient) aggregation, information such as class-conditional feature distributions or the amount of variance in the global population is never directly accessible to any of the participants. Also, performing dimensionality reduction may not always be a feasible option.

3.3.2 FDQM: Four-Dimensional Quantitative Measure for Statistical Heterogeneity

Guo et.al. [18] have come up with a four-dimensional metric that captures both the type and degree of Non-IID'ness in a Federated Setting. They identify four major causes of Non-IID'ness as Feature distribution skew ($p_i(X) \neq p_j(X)$), Label distribution skew ($p_i(Y) \neq p_j(Y)$), Concept drift (or "feature differency": $p_i(X|Y) \neq p_j(X|Y)$), and Quantity skew ($|D_i| \neq |D_j|$), and develop four distinct modeling approaches to quantify each component.

Label distribution and quantity skew are measured by modeling the distribution of label frequencies and data volumes, respectively, with a Dirichlet distribution, and inferring the degree of skew through certain properties of this distribution. The Dirichlet distribution is commonly used in non-parametric Bayesian statistics, and can be intuitively described as a distribution over probability mass functions. The probability density function of a K-dimensional Dirichlet distribution is given by:

$$f(x_1, \dots, x_K; \alpha_1, \dots, \alpha_K) = \frac{1}{|\mathcal{B}(\alpha)|} * \prod_{j=1}^K x_j^{\alpha_j-1} \quad (3.5)$$

, where $\alpha = (\alpha_1, \dots, \alpha_K)$, $\sum_{j=1}^K \alpha_j = 1$, and $x_j \in [0, 1] \ \forall j \in [K]$. The concentration parameter α determines how the probability mass is dispersed around the center of this k-dimensional sample space and can be used as a direct indicator of volatility. Guo et.al. [18] arithmetically derive an expression for α with regards to the mean, variance, and the number of clients, which is then used as a quantitative parameter of FDQM.

In order to quantify Feature distribution skew, they use an image noise estimation method based on eigenvalue decomposition, where the data is split into blocks of uniform size, and each block is modeled as a signal-noise mixture: $b = \hat{b} + \varepsilon$, where $\varepsilon \sim N(0, \sigma^2)$. Then, the standard deviation σ is isolated through eigenvalue decomposition and used as the Feature distribution skew parameter.

Finally, a Gaussian Mixture Model approach is used to quantify the distributional shift between clients' features within a given class/label.

4 Experiments and analysis

In this chapter, the experiments work conducted in the scope of this thesis is described. The experimental setup, including the used datasets, libraries and developed algorithms, are introduced. The explanation of the investigations are organized in three parts

4.1 Approach

Our experimental pipeline consists of three stages that build on each other: First, we generate a setting where we can experiment with different levels of heterogeneity in a reproducible manner. In order to do so, we take a partitioning approach (as opposed to sampling) in order to avoid any artificially induced correlation between the subsets (a proxy for clients in the Federated Setting). Multiple methods for generating heterogeneous subsets have been included in our experiments as a means to rule out any dependency on a specific partitioning method: we consider a probabilistic method, which uses the Dirichlet distribution to control distributional shift. We examine a third, quantity-based [19] method that randomly assigns regular portions of the original dataset to “clients”. In that case, the size of these portions determines the degree of heterogeneity.

The second stage consists of measuring the overall dissimilarity of the distributions, which will serve as a “universal truth” with regards to what we are trying to measure. We use goodness-of-fit tests and entropy-based divergence statistics to measure the pairwise dissimilarity between distributions. We simply use the average pairwise distance as a global statistic to describe the overall dissimilarity. We then go on to demonstrate the consistency of these methods, as well as their correlation with model accuracy. The latter part is achieved by training the same prediction model on different subsets generated in the first stage, applying the trained models to the same unseen test set, and comparing the prediction accuracy (measured simply as the ratio of correctly classified data instances) to our global statistic of overall distributional shift.

4.2 Datasets

- The MNIST (Modified National Institute of Standards and Technology) dataset is a widely used benchmark dataset for image classification tasks. It consists of 60000 28x28 grayscale images of handwritten digits from 0 to 9, with 6000 images per class. The dataset is split into 50,000 training images and 10,000 test images.
- The CIFAR-10 dataset is another widely used benchmark dataset for image classification tasks. It consists of 60000 32x32 color images in ten classes comprised of various animals and vehicles. The images are relatively low-resolution with rather complex features, which makes the classification task more challenging compared to MNIST.

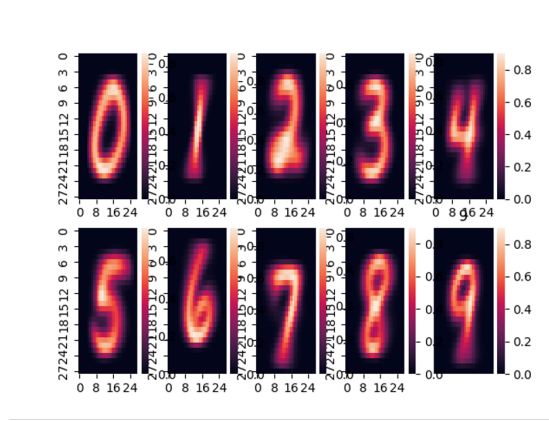


Figure 4.1: Average pixel data for each digit in the MNIST dataset.

4.3 Setup

The experiments were conducted in a Python environment and are publicly available at §. The datasets were imported from the torchvision library, the methods for statistical tests performed in the first stage of our pipeline were imported from scipy.stats. We used Tensorflow to implement machine learning-related tasks.

4.4 Part I: Simulating a Controlled, Heterogeneous Federated Learning Environment

When it comes to testing hypotheses w.r.t. Federated Learning, we are faced with the unique challenge that the idea of a dataset drawn from an actual Federated environment contradicts the founding principle of privacy and limited data sharing. This limits our options in testing with Federated, real-world data. While synthetic data could alleviate this problem, it would also make the results obtained with it relative to the specific setting, so direct comparisons with traditional approaches would be difficult. At the same time, we need more control over statistical heterogeneity in these test environments to be able to study its correlation with secondary factors.

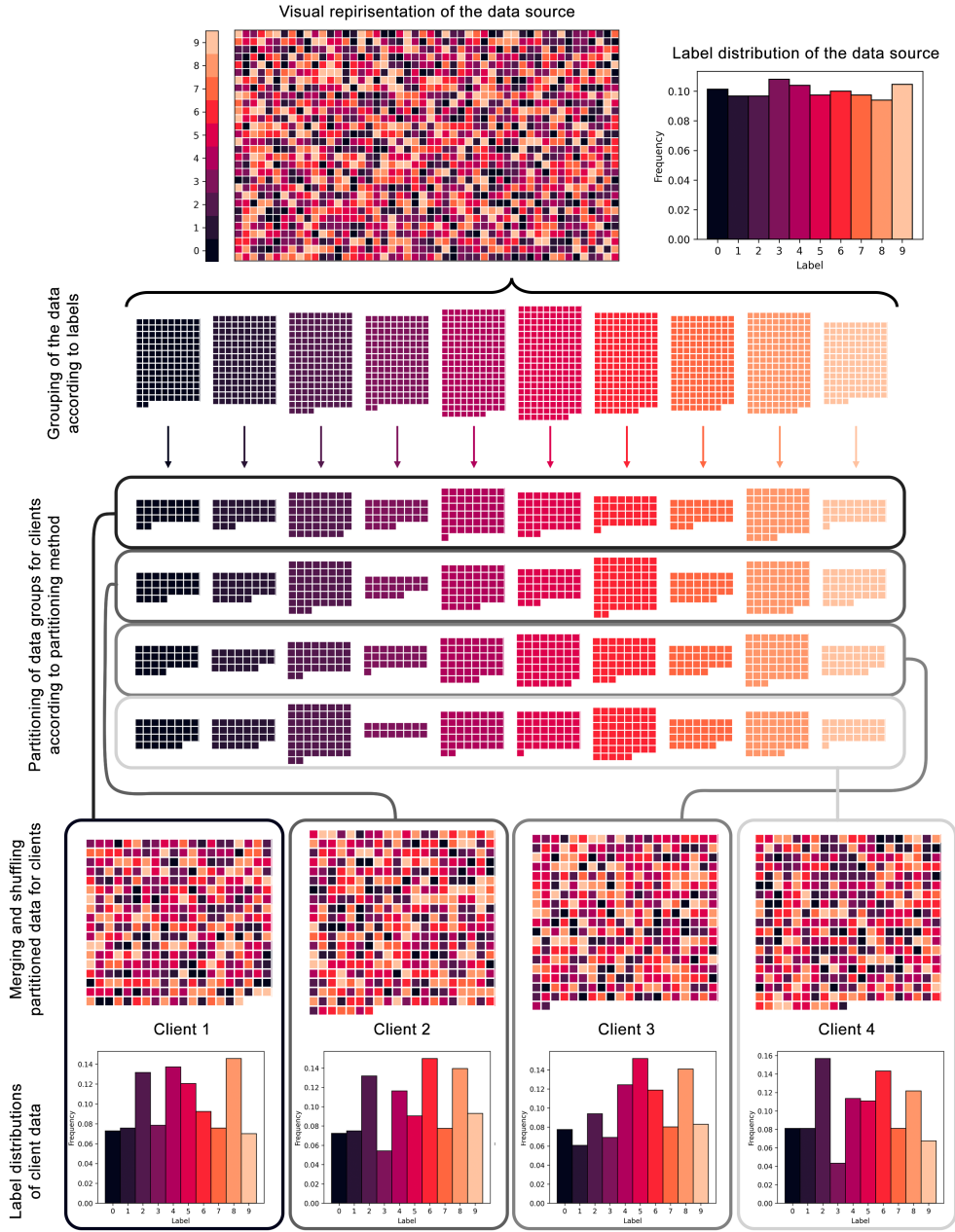


Figure 4.2: Visual representation of label-driven data partitioning to multiple clients. In this example data consists of 10 labels (similar to the MNIST and CIFAR10 data-sets) which is partitioned homogeneously to four clients.

One common way of simulating heterogeneous environments in a controlled and reproducible manner is via partitioning, which has been particularly popular in Federated Learning research related to convergence analysis in Non-IID settings. Here, a real-world benchmarking dataset such as MNIST is split into multiple disjoint subsets under various strategies that dictate the distributional shift between them. There are a couple of partitioning strategies that are adequate for our focus on label distribution skew, and the following chapters are devoted to their analysis. .

4.4.1 Homogeneous Partitioning

As discussed in detail in previous chapters, the ideal situation in a Horizontal Federated Learning scenario would be one where local data distributions are (near-) identical, such that the learned global model would generalize perfectly to any client domain. This ideal state of (near-) perfect statistical homogeneity constitutes our point of origin to measure other environments against, so we need to simulate this as well.

The heterogeneous partitioning methods that we will discuss shortly can also be used to create homogeneous datasets at their extremes, but we have included a third method for homogeneous partitioning for the sake of keeping our “control case” clean and isolated. In particular, we have used a built-in stratification method from the scikit-learn library originally meant for cross-validation tasks to split the MNIST and CIFAR10 datasets into 10 subsets such that the label frequencies within each subset approximately follow the same distribution. The stratified subsets are then stored as a list structure containing the corresponding indices which can be accessed through the dictionary returned by the function.

The label frequency distribution in the original MNIST dataset is approximately unifor. The results of our homogeneous partitioning method can be seen in 4.2.

4.4.2 Quantity-Based Partitioning

This partitioning strategy was first used in a study by McMahan et.al. [20], where FedAvg’s behavior under unbalanced and Non-IID data was empirically analyzed. This is a rather heuristic partitioning approach where the data is split into chunks with varying sizes, and grouped to form equally sized subsets representing clients. (As our focus lies on label distribution, quantity skew must remain fixed, so we have kept the sizes of the subsets homogeneous in all partitioning methods.)

Our algorithmic implementation of this strategy consists of a few simple steps: We first sort the training set by labels such that we obtain a series of “pure chunks” and store the corresponding index mapping in an array structure. Then we randomly split this array into $n * \alpha$ batches of size $\frac{N}{n * \alpha}$, where n denotes the number of clients

and N the number of data points in the dataset. These mostly pure batches are then reshuffled (without changing their inner structure) and grouped by simple random permutation and splitting to form n different subsets. So, the quantity α^{-1} controls the size of the batches, and thereby the degree of skewedness (At the extreme case where $\alpha = \frac{N}{n}$, each chunk consists of a single datapoint and the entire process degenerates into uniform reshuffling with approximately homogeneous results. The other extreme is when we have $\alpha = 1$ and the chunks, still mostly pure depending on the proportion of client- vs. distinct label count, are at their maximum size under a fixed n . This leads to each subset/client containing only some of the labels, which is referred to as the “pathological” non-IID scenario.).

Below 4.3 we see the label frequency distributions obtained after eight runs of quantity-based partitioning with $\alpha = [1, 10, 30, 60, 150, 300, 3000, 6000]$ and $n = 10$, represented as heatmaps where each row represents the label distribution within a single subset/client.

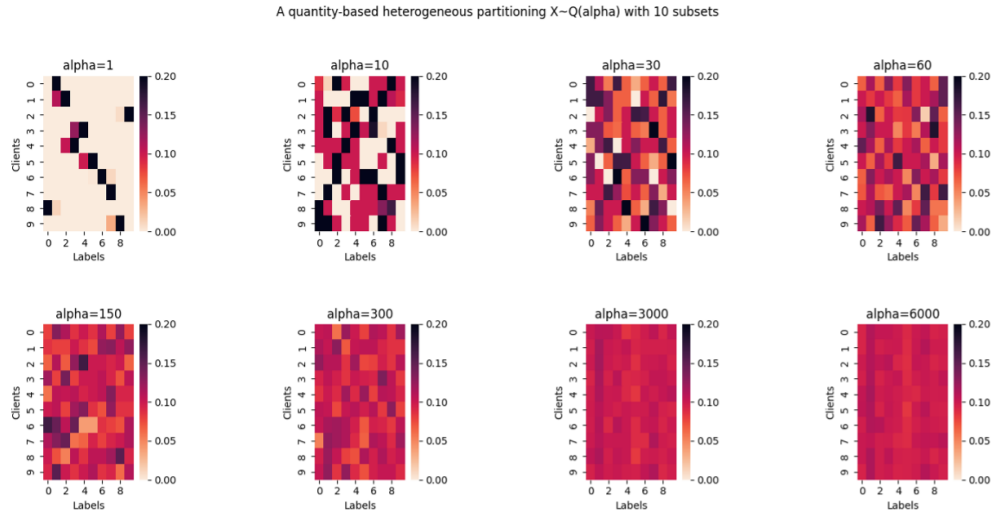


Figure 4.3: Label heterogeneity in quantity based partitioned data with varying α value.

As predicted, $\alpha = 1$ has led to the “pathological case” where only one or two out of ten unique labels are present in each subset. This setting could be used as a baseline to analyze worst-case scenarios. For larger α , we see a clear shift towards heterogeneity, which, for our purposes, validates this partitioning strategy in a practical sense.

4.4.3 Distribution-Based Partitioning: Dirichlet

The Dirichlet distribution is a multivariate and continuous probability distribution commonly used in non-parametric Bayesian statistics. A Dirichlet-distributed K -dimensional random vector $X = (X_1, \dots, X_K)$ has its support over the $(K - 1)$ -dimensional simplex, i.e., the random variables $X_1, \dots, X_K$ have nonzero values only under the condition that: $\forall j \in [K] : X_j \in (0, 1)$ and $\sum_{j=1}^K X_j = 1$. Thus, any legal value of the random vector X can be interpreted as a probability distribution over a fixed number of outcomes (Multinomial Distribution). The Dirichlet distribution then assigns a likelihood to every legal outcome for $X = (X_1, \dots, X_K)$.

Let's depart from our actual experimental setting for a while and assume we have a simpler dataset with the label space $Y = \{A, B, C\}$, which we want to partition into five subsets with the same objective and expected results as in the quantity-based approach. We could model the process of distributing the data instances among the subsets as a random process where for each label, let's say A , the data points containing the label A are assigned to one of the five clients with the probabilities $\theta_A = (\theta_A^1, \theta_A^2, \theta_A^3, \theta_A^4, \theta_A^5)$, where $\sum_{j=1}^5 \theta_A^j = 1$. So, we will obtain three five-dimensional probability vectors $\theta_A, \theta_B, \theta_C$ such that $\theta_{L_i} := \Pr[S(x) = L \mid x \in D_i] \forall L \in \{A, B, C\}, i \in [5]$, where $S : X \rightarrow Y$ denotes the supervisory mechanism mapping datapoints to their true labels and D_i denotes the i -th subset. If we now return to our original problem statement, the overall dissimilarity among these random vectors $\theta_A, \theta_B, \theta_C$ is exactly what we want to model, and this can be done very elegantly with Dirichlet-sampling, as we will see shortly. But let us first have a look at the probability density function of a Dirichlet-distributed random variable of order K :

$$X = (X_1, \dots, X_K), \alpha = (\alpha_1, \dots, \alpha_K); X \sim \text{Dir}_K(\alpha) \Rightarrow f(x; \alpha) = \frac{1}{B(\alpha)} * \prod_{i=1}^K x_i^{\alpha_i - 1}$$

, where the beta-function $B(\alpha) = \frac{\prod_{j=1}^K \Gamma(\alpha_j)}{\Gamma(\sum_{i=1}^K \alpha_i)}$ serves as a normalizing constant (i.e., to ensure that $\int f(x; \alpha) dx = 1$).

As evident from the above probability density function, the K -dimensional parameter vector $\alpha = (\alpha_1, \dots, \alpha_K)$ directly determines the likelihood assigned to various random events ($X_1 = x_1, \dots, X_K = x_K$) or, in more general terms, how the probability mass is dispersed around the center of the K -dimensional simplex. Although it is not possible to visualize higher-dimensional surfaces directly, we can use the Beta distribution (which corresponds to a Dirichlet distribution of order 2), or Dirichlet distribution over a three-dimensional simplex in order to demonstrate how different values of the concentration parameter α lead to different characteristics in probability mass distribution.

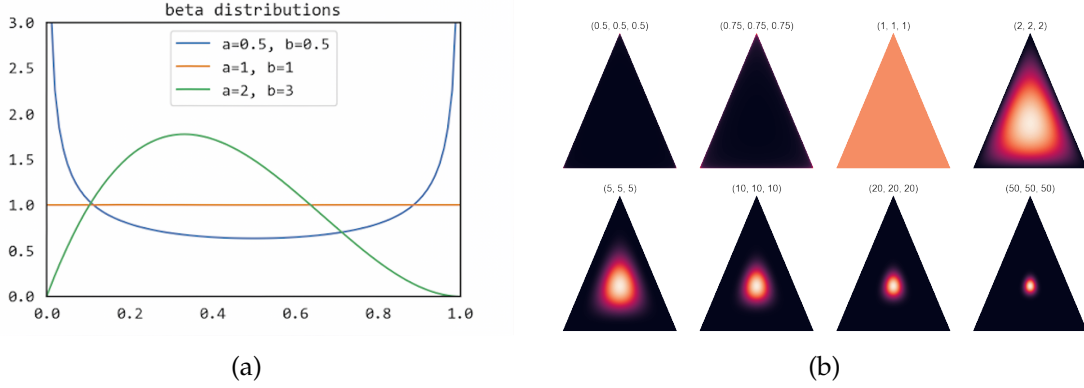


Figure 4.4: (left) Beta probability density functions with various a and b parameters. (right) Dirichlet probability mass distributions over a 3D simplex with equally varying α parameters.

When there is no prior information available on the relative likelihoods among the categories (e.g., the probability of selecting a data instance from a certain class into one of the five groups in the previous example), the α_i are usually assumed to be equal. This special case is referred to as the symmetrical Dirichlet distribution. We will also assume a symmetrical Dirichlet distribution in our modeling since the distributional relationship between the labels themselves isn't of much interest to us, and also because this information not being available a-priori is a realistic one. So, we will only look at symmetrical distributions to define how different values of the concentration parameter α effects the probability mass distribution. In general, we identify three distinct cases:

1. with the parameters $a = b = 1$ in Figure 4.4a, where the probability mass is concentrated around the values 0.0 and 1.0, as well as in the three-dimensional Dirichlet distribution with $\alpha = 0.85$, where the probability mass is concentrated around the edges of the simplex. The label distribution under $\text{Dir}(\alpha < 1)$ can be compared to the “pathological case” in quantity-based partitioning.
2. $\alpha = 1$: This is the homogeneous case (in terms of our model's preference towards various values of $X = (X_1, \dots, X_K)$, and not the distribution of X itself), where each point on the support simplex is equally likely. The label distribution under $\text{Dir}(\alpha = 1)$ could be described as “perfect randomness”.
3. $\alpha > 1$: As can be concluded per simple inspection on the probability density function of the Dirichlet distribution, under $\alpha > 1$ with positive-valued exponents over $x_j \in (0,1) \ \forall j \in [K]$, values of the probability vector X which maximize

$f(x; \alpha)$ will increasingly resemble a uniform distribution, as we keep increasing the concentration parameter α .

In summary, we have a perfect tool for modeling label distribution skew through the Dirichlet distribution thanks to its interplay with multinomial distributions -which is basically the only way of modeling label frequencies probabilistically- and the ease of specifying the degree of skewedness through the concentration parameter α .

Now that we have clarified why the Dirichlet distribution is uniquely suited for our modeling purposes, we can explain how a partitioning strategy based on this works: As with the previous approach, we start by sorting the dataset by labels into “pure” sets. Then, for each label $k \in K$, we generate a random vector $p_k \sim \text{Dir}(\alpha)$ that is randomly sampled from a Dirichlet distribution simulated with the pre-specified concentration parameter α . The individual components of these random vectors $p_{k,i}$ (where $i \in [n]$ and n denotes the number of clients/subsets) are then normalized ($p_{k,i} \leftarrow n * p_{k,i}$ and $n = \sum_{k \in K} p_{k,i} \forall i \in [n]$) to prevent quantity skew. These normalized vectors p'_k are then used to split the pure-label subsets into chunks of corresponding sizes.

We have performed this partitioning method on the MNIST dataset with $n = 10$ clients and $\alpha \in 0.5, 5, 10, 50, 100, 1000, 5000, 10000$ for the purpose of providing a simple visual validation, which can be seen below.

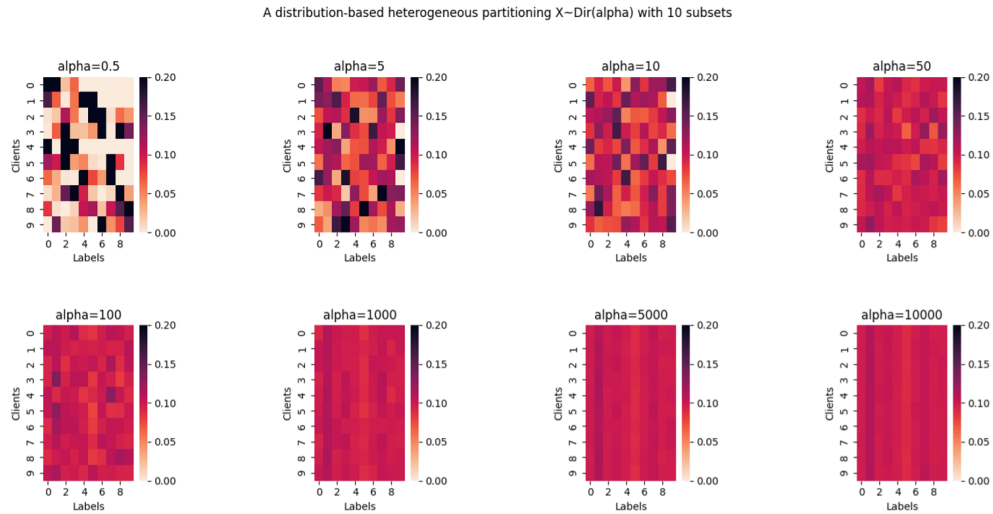


Figure 4.5: Label heterogeneity in distribution based partitioned data with varying α value.

4.5 Part II: Measuring Statistical Similarity of Samples

In the previous section, we have discussed how we simulate Federated datasets with varying degrees of divergence between label distributions. So, we can start returning our attention back to our primary problem of quantifying the mean distance between a set of data distributions. For this purpose, we will first look at the more fundamental question of how we can quantify the distributional distance between two datasets. Since information on the local distributions isn't directly available in a Federated setting, it has to be said that none of the measures that we'll be discussing here would be applicable as a global heterogeneity metric. We instead intend to establish a "ground truth" on the degree of heterogeneity among the set distributions based by using well-defined and well-studied statistical methods, so that later evaluations of Non-IID'ness can be based on these earlier results. To this end, we will look at multiple strategies for measuring the (dis-) similarity of two sets including nonparametric statistical tests, information theoretical metrics, and kernel distances, as well as a few simple heuristics for choosing an appropriate similarity measure. But let us clarify our problem definition for this section:

Definition 5: (Dissimilarity of Sets) Given two higher-dimensional sets $D_1, D_2 \subseteq X$, we want to define a function $d : X \times X \rightarrow \mathbb{R}$ that quantifies the dissimilarity (or distance) between these two sets. In general, we can expect the following properties from a dissimilarity function d for all $x, y \in X$ [21]:

- Non-negativity: $d(x, y) \geq 0$
- Symmetry: $d(x, y) = d(y, x)$
- Reflexivity: $d(x, x) = 0$

Such a dissimilarity measure could be based on some notion of point-wise geometric distance (like the Euclidian distance), or, we could interpret X and Y as manifestations of two different random quantities P_X, P_Y and use their statistical distance as a dissimilarity measure. There are many different versions of both point-wise and statistical distances, as we will see later.

After the distance function d has been established, we can form a summary statistic for more than two sets by simply computing the mean distance between pairs of sets:

$$D_n = \frac{1}{|\{(X_i, X_j) : i \neq j; i, j \in [n]\}|} * \sum_{k=1}^{n-1} d(X_1, X_2) \quad (4.1)$$

Definition 6: (Similarity of Sets) We can convert a given well-defined dissimilarity metric $d : X \times X \rightarrow \mathbb{R}$ into a similarity metric $s : X \times X \rightarrow \mathbb{R}$ by simply applying a

real-valued monotonically decreasing function f , e.g., $s(x, y) = f(d(x, y)) = e^{\frac{-d(x, y)}{\sigma_X}}$ or $s(x, y) = \frac{1}{1+d(x, y)}$. [33,34]

4.5.1 Vector-based Dissimilarity

Dissimilarity between two sets of same-dimensional data could be measured through the mean distance between their elements represented as vectors in a higher-dimensional metric space, e.g., $X = \{x_i \in \mathbb{R}^d\}_{i=1}^n, Y = \{y_j \in \mathbb{R}^d\}_{j=1}^m$. A very common notion of metric distance is provided by the Minkowski family, which can be described as a generalization of the Euclidian distance:

$$M_d(x, y) = \left(\sum_{i=1}^d |x_i - y_i|^d \right)^{1/d}, \text{ where } x, y \in \mathbb{R}^d \quad (4.2)$$

This notion of distance can also be used to measure the statistical distance between two probability density functions $p(x), q(x)$ defined on the same domain X (statistical Minkowski distance, [22]):

$$M_\alpha(p, q) = \left(\int_{x \in X} |p(x) - q(x)|^\alpha dx \right)^{\frac{1}{\alpha}}, \text{ where } \alpha \geq 1 \quad (4.3)$$

In our case, we could represent the categorical distribution of labels as a vector $p = (p_1, \dots, p_K)$ where $\sum_{i=1}^K p_i = 1, p_i \geq 0 \forall i \in [K]$, and use the Minkowski distance to quantify their dissimilarity. Another vector-based distance metric commonly used in data mining and information retrieval is the cosine distance, which is based on the angle between two vectors:

$$d_{\cos}(x, y) = 1 - s_{\cos}(x, y) = 1 - \cos(\theta_{xy}) = 1 - \frac{x \cdot y}{\|x\|_2 \|y\|_2} = 1 - \frac{\sum_{i=1}^d x_i y_i}{\sqrt{\sum_{i=1}^d x_i^2} \sqrt{\sum_{i=1}^d y_i^2}} \quad (4.4)$$

In the case of proportional vectors $x \parallel y$ we obtain $d_{\cos}(x, y) = 1 - \cos(0) = 0$ and for orthogonal vectors $x \perp y$, $d_{\cos}(x, y) = 1 - \cos(\frac{\pi}{2}) = 1$.

4.5.2 Statistical Distance

Statistical distance refers to the dissimilarity between two random objects and is mostly measured through a direct comparison between the probability density functions of these two random quantities. Commonly used statistical distance metrics can be broadly categorized into two groups as “metrics” and “divergences”. Metrics can

be regarded as an extension of geometrical distance to probabilistic functions, much like the previously mentioned statistical Minkowski distance, where divergences are mostly based on information-theoretical concepts like entropy. In realistic data analysis scenarios, a-priori information on probability densities is in general not available, but statistical distances can still be used to quantify the dissimilarity between empirical quantities such as label distribution. Here, we will look at some examples of statistical distance.

Mahalanobis Distance

Originally developed to measure the distance between a point and a probability distribution, the Mahalanobis distance is often used in statistical analysis. For a data point $x \in \mathbb{R}^d$ and a random variable P with the first-order moment μ_P , the Mahalanobis distance is defined as [cite]:

$$d_M(x, \mu_P) = \sqrt{(x - \mu_P)^T S^{-1} (x - \mu_P)} \quad (4.5)$$

where S is the positive-definite covariance matrix: $S = \mathbb{E}[(P - \mu_P)(P - \mu_P)^T]$. It can be extended to compare distributions based on their means under the assumption that their covariance matrices are identical [cite]:

$$d_M(\mu_1, \mu_2) = \sqrt{(\mu_1 - \mu_2)^T S^{-1} (\mu_1 - \mu_2)} \quad (4.6)$$

The population means $\mu_{1,2} = \mathbb{E}[P_{1,2}]$ can be substituted with sample means $\hat{\mu}_{1,2} = \bar{x}_{1,2}$, $x_{1,2} \sim P_{1,2}$, and the covariance matrix can be substituted with $\Sigma = \frac{1}{N-1} * \sum_{i=1}^N (x_i - \bar{x})(x_i - \bar{x})^T$ to obtain an empirical estimate of dissimilarity between two distributions:

$$d_M(x_1, x_2) = \sqrt{(x_1 - x_2)^T \Sigma^{-1} (x_1 - x_2)} \quad (4.7)$$

The Mahalanobis distance is especially adequate for comparing higher-dimensional numerical datasets with a shared covariance structure.

Wasserstein Distance

Wasserstein distance (also known as earth mover's distance) defines the dissimilarity between two probability distributions or datasets as the minimum amount of work required to transform them into one another (optimal transport problem), where the amount of work is measured as the product of "weight differential" between them, times the transportation cost. For two probability distributions X and Y , it can be

written as [23], [24]:

$$W_p = \left(\inf \mathbb{E} [|X - Y|^p] \right)^{1/p} = \left(\int_{\mathbb{R}} |F_X(t) - F_Y(t)|^p dt \right)^{1/p}, \quad (4.8)$$

where $F_X(t) = \Pr[X \leq t]$.

Kolmogorov-Smirnov (KS) Statistic

The Kolmogorov-Smirnov statistic is based on the maximum distance between empirical cumulative distribution functions. It is commonly used in goodness-of-fit and hypothesis tests, and can be defined for two random samples $\{x_i\}_{i=1}^n \sim X$, $\{y_j\}_{j=1}^m \sim Y$ as:

$$KS(X, Y) = \sup_{t \in \mathbb{R}} |F_X(t) - F_Y(t)|, \quad (4.9)$$

where F_X and F_Y refer to (empirical) cumulative distribution functions: $F_X(t) = \sum_{i=1}^n 1_{x_i \leq t}$ and $F_Y(t) = \sum_{j=1}^m 1_{y_j \leq t}$. The KS statistic's use is restricted to one-dimensional distributions since a generalization into higher dimensions is regarded as a non-trivial task due to the ambiguities involved in evaluating a supremum over multiple dimensions [25]. The KS statistic can be preferred over other alternatives in situations where ease of evaluation is a major concern or when no prior information on the distributions is available beforehand. Though one could say that it has less representation power than some other metrics since the shape of the distributions isn't considered.

Kullback-Leibler (KL) Divergence

The KL divergence for two random variables P, Q defined on the same domain X can be defined as [32,40]:

$$\begin{aligned} d_{KL}(P, Q) &= \sum_{x \in X} p(x) * \log \left(\frac{p(x)}{q(x)} \right) = \mathbb{E}_{x \sim P} \left[\log \left(\frac{p(x)}{q(x)} \right) \right] \quad (\text{discrete case}) \quad (4.10) \\ d_{KL}(P, Q) &= \int_{-\infty}^{\infty} p(x) * \log \left(\frac{p(x)}{q(x)} \right) = \mathbb{E}_{x \sim P} \left[\log \left(\frac{p(x)}{q(x)} \right) \right] \quad (\text{continuous case}) \end{aligned} \quad (4.11)$$

It can be expressed in terms of entropy as:

$$\begin{aligned} d_{KL}(P, Q) &= \sum_{x \in X} p(x) * \log \left(\frac{p(x)}{q(x)} \right) = - \sum_{x \in X} p(x) * \log \left(\frac{q(x)}{p(x)} \right) \\ &= - \sum_{x \in X} p(x) * \log(q(x)) + \sum_{x \in X} p(x) * \log(p(x)) = H(P, Q) - H(P), \end{aligned} \quad (4.12)$$

where $H(P, Q)$ refers to the cross-entropy between P and Q , and $H(P)$ is the entropy of P . It can be interpreted as the information that P carries on Q , and is often referred to as “relative entropy” and “information gain”. It should be noted that the KL distance isn’t symmetric ($d_{KL}(P, Q) = H(P, Q) - H(P)$ isn’t necessarily equal to $d_{KL}(Q, P) = H(Q, P) - H(Q)$), thus it isn’t a metric quantity. In situations where the symmetry property is a necessary requirement, we could use the Jeffrey distance, which is a simple symmetrical extension of the Kullback-Leibler divergence:

$$d_J = d_{KL}(P, Q) + d_{KL}(Q, P)$$

Gini Index

The Gini Index is often used in economics to describe income equality and is not a dissimilarity measure by itself. But it very powerful in describing the equality of populations, so we have included it in our experiments where we our main point of reference (label distribution in benchmarking sets MNIST and CIFAR10) also follows an approximately uniform distribution. We have used a simple implementation based on the following formula:

$$Gini(X) = \frac{1}{|X|^2 * \hat{\mu}_X} \sum_{i=1}^{|X|} \sum_{j=i+1}^{|X|} |x_i - x_j|, \text{ where } x_i, x_j \in X$$

4.5.3 Kernel Distances

Vector-based distance metrics in general, and a majority of the statistical distance measures we have discussed work very well with univariate distributions and are cut perfectly for the task of comparing label distributions or measuring loss in the machine learning context. Though when it comes to comparing the feature space, these notions of dissimilarity often fall short of capturing the information present in higher-dimensional data in its entire complexity. This is especially relevant for very high-dimensional classification problems, where Kernel methods (e.g., Support Vector Machines, Principal Component Analysis) have emerged as the prominent solution approach.

The basic principle behind kernel methods is the projection of data into a higher-dimensional feature space, where patterns can be identified more easily. The similarity between data instances can then be expressed through a dot product between their higher-dimensional representations.

4.5.4 Results

In 3.1., we presented some partitioning strategies with which a Federated Learning environment can be simulated at different degrees of statistical heterogeneity. But, apart from a graphical density representation, we haven't yet examined how subsets produced with these partitioning strategies at varying hyperparameter values behave in terms of statistical heterogeneity. In order to close this gap, and also to obtain reference values for the statistical distance measures that we'll be using, we have conducted a simple experiment with MNIST and CIFAR10 datasets. We have chosen the Kullback-Leibler divergence, Kolmogorov-Smirnov statistic, Wasserstein distance, and Gini index as our dissimilarity measures with the intention of capturing different notions of distance/dissimilarity.

Quantity-based Partitioning

The MNIST dataset contains 60000 28×28 greyscale images of handwritten digits. We have generated 10 disjoint, equal sized subsets (which ultimately represents clients in a Federated setting) by applying the quantity-based partitioning approach over 40 iterations where the "granularity" parameter α was drawn from the set of positive divisors for the maximum subset size.

Since we are primarily interested in label distribution skew, each subset was mapped to a 10-dimensional vector $p_k^\alpha = (p_{k,0}^\alpha, \dots, p_{k,9}^\alpha)$. Then, for each collection of subsets $\{p_k^\alpha\}_{k=1}^{10}$ generated under a fixed granularity parameter $\alpha \in \{n \in \mathbb{Z} \mid 6000 \equiv 0 \pmod{n}\}$, we have computed the mean divergence from the global parameter distribution $p_0 = (p_{0,0}, \dots, p_{0,9})$ using all four of the aforementioned statistics. This process was repeated over 10 epochs with different random seeds, and the average of mean dissimilarity statistics with respects to the null distribution were plotted against the α -values. The variability within the mean dissimilarity results over multiple runs was simply measured through standard deviation and represented as "uncertainty regions" around the mean, as can be seen in figures 4.6.

Apart from some volatility in Kolmogorov-Smirnov statistics, the results show a similar overall trend: At $\alpha = 1$ (marked as "0" on the x-axis), we observe what we have previously called "the pathological case", where the subsets contain only one or two out of ten possible labels. Values obtained for each statistic at $\alpha = 1$ can be regarded as a fuzzy upper bound on the degree of heterogeneity that is achievable with our current methods. (Also, we partially observe values quite close to the actual upper bound as can be seen in results featuring the Kolmogorov-Smirnov statistic, which means we can approximate perfect heterogeneity with quantity-based partitioning strategy.) Then, approximately up to $\alpha = 3000$, we see an exponential decrease in mean

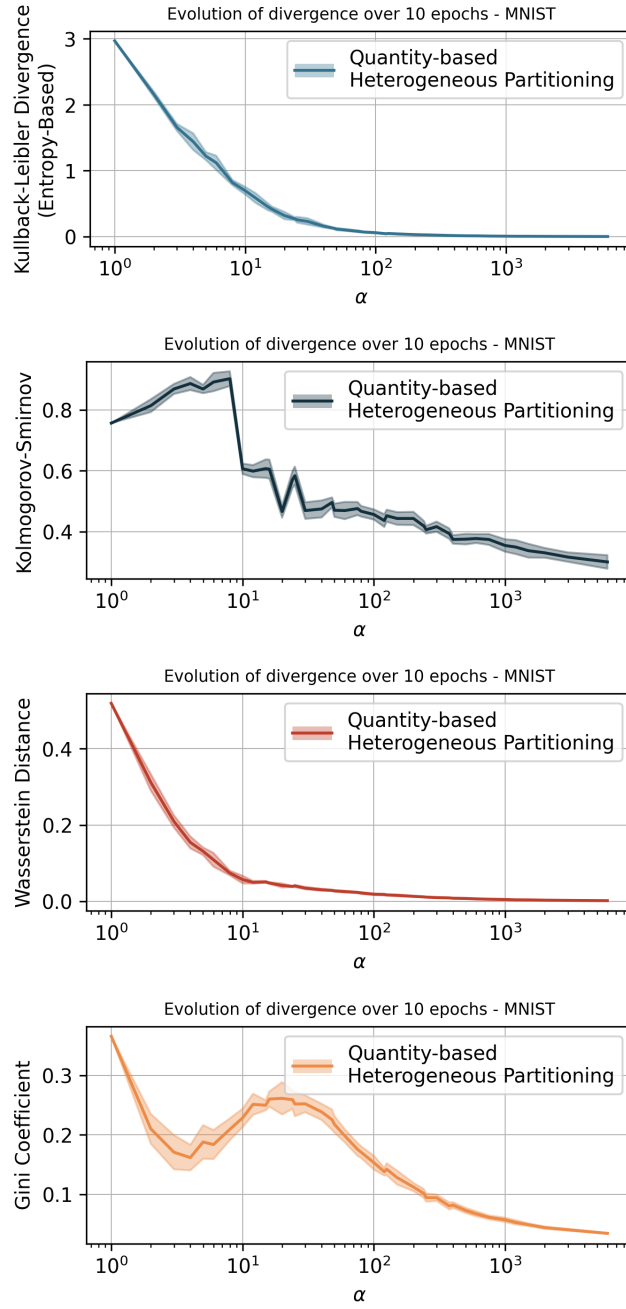


Figure 4.6: The heterogeneity of quantity-based heterogeneous partitioned MNIST data, determined with four dissimilarity measures. Top-down: Kullback-Leibler divergence, Kolmogorov-Smirnov statistic, Wasserstein distance, and Gini index

values. Whereas the observed monotonic decrease in heterogeneity does confirm our methodology, geometric properties of the curve are mostly the result of inhomogeneous measurement density (α is drawn from the set of positive divisors of 6000), and we don't suspect them to carry further meaning.

At $\alpha = 6000$ (marked as "60" on the x-axis) [§ needs fixing] we observe the marginal case of maximum granularity where the datapoints are uniformly assigned to 10 subsets at random. In this case we expect the subsets' label distributions to be approximately the same as the original distribution. This is confirmed by values close to zero for all four dissimilarity measures. Overall, we can conclude that the quantity-based partitioning strategy behaves as expected.

Since both our partitioning and dissimilarity measurement strategies are targeted at label distributions, other properties of the data used in the experiment aren't relevant for the results obtained in this section, thus, we don't expect to see a significant divergence in results obtained with other datasets of similar size and label distribution. However, the third stage of our experiments uses the CIFAR10 dataset in addition to MNIST, so we have validated our partitioning strategies on the basis of CIFAR10 as well, as can be seen in 4.7.

Now that we have confirmed the effectiveness of this method, we can turn our attention to a more detailed analysis of the differences and commonalities among these results. The first question that arises is the extent to which these four metrics exhibit similar behavior, and which are more robust to small changes in the data (MNIST and CIFAR10 have different sample sizes, and the label distributions aren't perfectly uniform so that small but measurable differences can be expected.) This question can be answered through a simple comparison between the two set of results we've obtained with MNIST and CIFAR10 datasets: In the case of Kullback-Leibler divergence our measurements almost perfectly match each other, whereas with Wasserstein Distance we observe the same high-level behavior, but with a slightly different magnitude. The KS-statistic and Gini-Coefficient on the other hand show significant difference across datasets, as well as higher variance compared to the other two metrics. Furthermore, we see some relatively complicated monotony behavior in these results, which is not expected under our current method and can be interpreted as an inconsistency. Thus, we will primarily use Kullback-Leibler divergence and Wasserstein-distance to specify dissimilarity.

Another interesting question we could ask is whether it is possible to identify a critical point or region where the measured statistical distances undergo a significant shift in behavior. Although this is hard to answer in general terms, we do see some

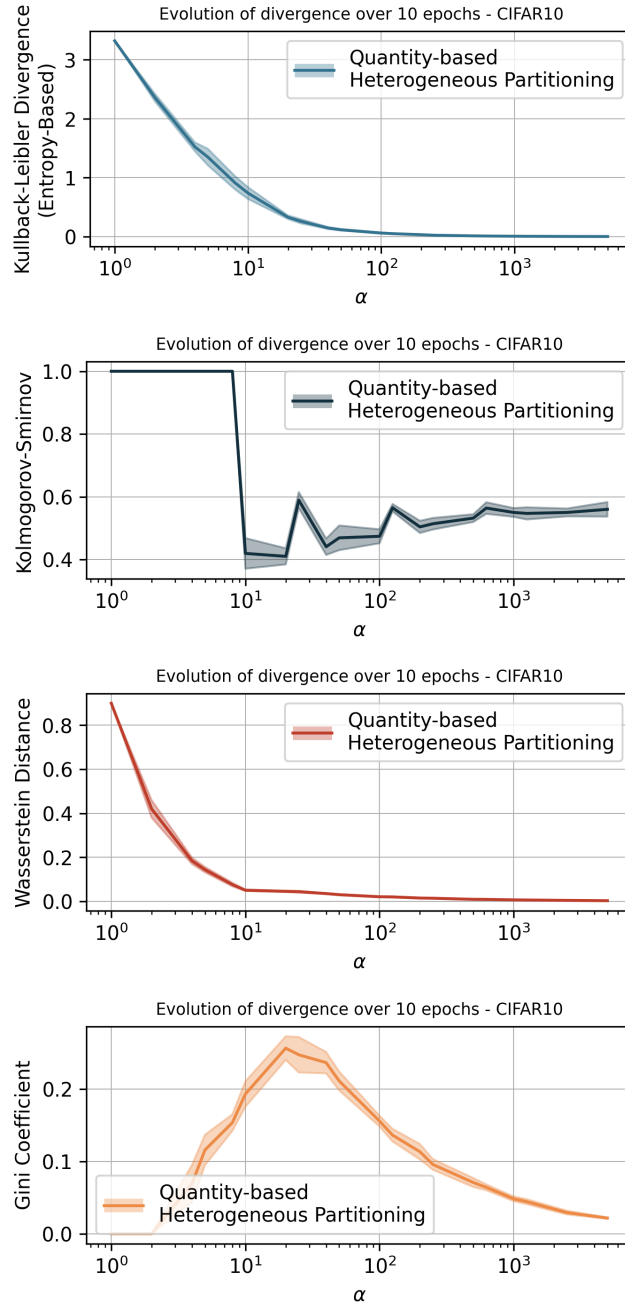


Figure 4.7: The heterogeneity of quantity-based heterogeneous partitioned CIFAR10 data, determined with four dissimilarity measures. Top-down: Kullback-Leibler divergence, Kolmogorov-Smirnov statistic, Wasserstein distance, and Gini index

Distribution-based Partitioning

To evaluate the distribution-based approach, we partitioned the MNIST dataset into 10 subsets and measured their divergence from the original dataset w.r.t. label distributions. Figures [fig] show the average test statistics obtained with concentration parameter values $\alpha \in [0.1, 20]$ (step size 0.1) over 10 runs.

Compared to the quantity-based method, the results for the distribution-based approach are more consistent across different metrics, as can be seen in Figure [fig]. Around $\alpha = 1$, we see an “inverted elbow”- the decrease in measured dissimilarities past this point is diminishing compared to the ones prior to $\alpha = 1$. This reflects the property of the Dirichlet distribution that concentration parameter values $\alpha < 1$ lead to a higher “ranking” of sparse distributions, which means in our case that that subsets will likely only have access to a small portion of the label space. As, again, predicted by properties of the Dirichlet distribution, we see a convergence towards homogeneity, as α approaches zero.

The results obtained with CIFAR10 almost perfectly match those obtained with MNIST, only with the KS-statistic showing increased uncertainty, which can be linked to the reduced dataset size.

Another remarkable difference to the quantity-based method is the improved consistency of results using the Kolmogorov-Smirnov statistic and Gini Coefficient. In addition to being consistent in overall behavior with the other two metrics, they show little variability across the two sets of measurements using MNIST and CIFAR10. More importantly, their values seem to align with the expected behavior of a distribution-based partitioning (KS/Gini=1 $\rightarrow$ perfect inequality; KS/Gini=0 $\rightarrow$ perfect equality). This elicits the question whether the inconsistent monotony behavior we have observed in the quantity-based approach was the result of an undefined local behavior of the algorithm itself, or the result of an algorithmic error. We did not investigate this question further because access to a single reliable partitioning strategy suffices for our purposes.

Furthermore, the quantity-based approach is heavily dependent on the specific properties of the data such as the number of distinct classes, label distribution, number of clients, and the sample size, all of which must be taken into account while implementing the algorithm. The distribution-based approach on the other hand isn’t at all dependent in its formulation on data properties, offers finer-grained control over the degree of heterogeneity (legal range of α represents a continuum), and is generally more robust (reduced variance) than the quantity-based approach. Thus, we will exclusively use the distribution-based approach in the final stage of our experiments.

As previously stated, another motivation for conducting statistical distance measurements on partitioned data was to obtain reference values our chosen set of metrics with

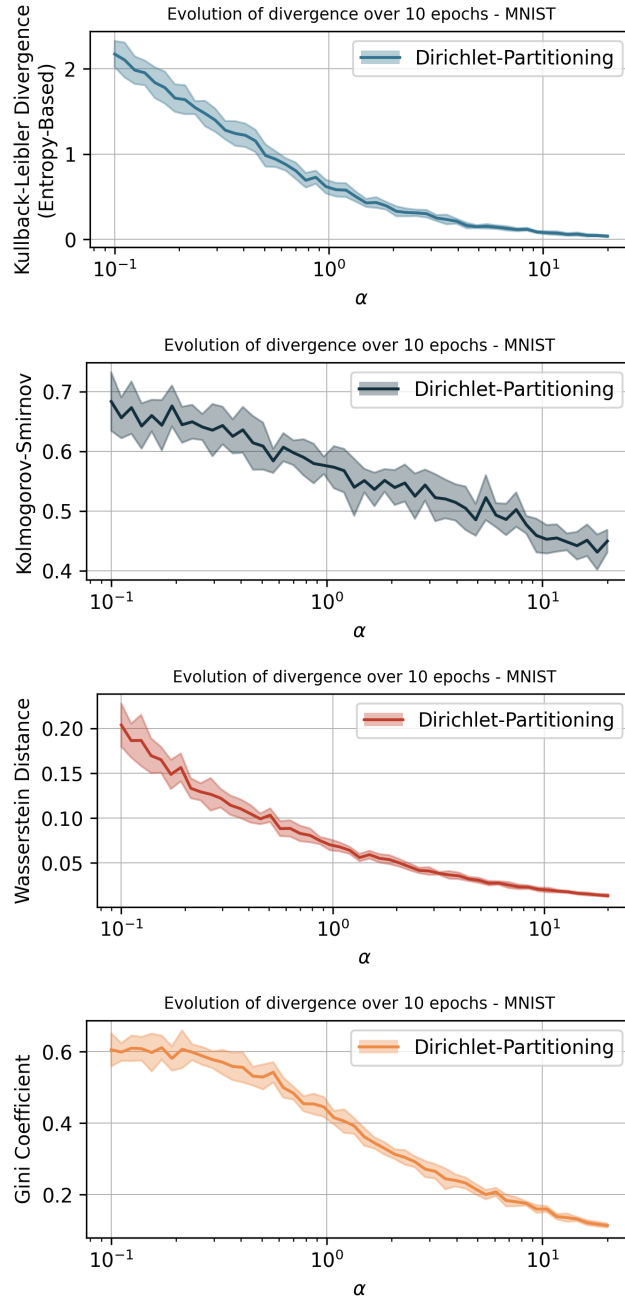


Figure 4.8: The heterogeneity of Dirichlet partitioned MNIST data, determined with four dissimilarity measures. Top-down: Kullback-Leibler divergence, Kolmogorov-Smirnov statistic, Wasserstein distance, and Gini index

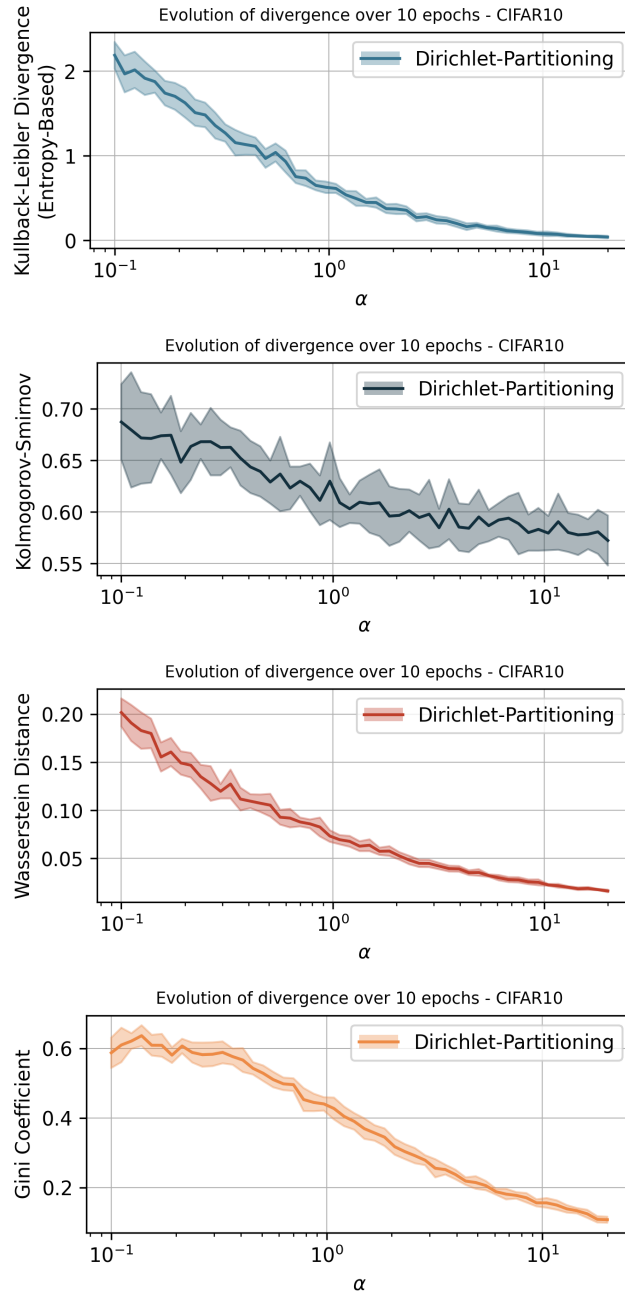


Figure 4.9: The heterogeneity of Dirichlet partitioned CIFAR10 data, determined with four dissimilarity measures. Top-down: Kullback-Leibler divergence, Kolmogorov-Smirnov statistic, Wasserstein distance, and Gini index

regards to hyperparameter values (α). This is especially important for Kullback-Leibler divergence and Wasserstein distance, since these aren't normalized metrics and require reference values for interpretation. Thus, we have included a list of reference values for various α , as can be seen in Table 4.1.

Dirichlet Partitioning ($\alpha = 1$)	MNIST	CIFAR10
Kullback-Leibler Divergence	0.637 ± 0.025	0.639 ± 0.031
Kolmogorov-Smirnov	0.581 ± 0.011	0.608 ± 0.007
Wasserstein Distance	0.068 ± 0.002	0.078 ± 0.002
Gini Coefficient	0.431 ± 0.008	0.423 ± 0.007

Table 4.1: Reference heterogeneity values for Dirichlet partitioning with $\alpha = 1$.

4.6 Part III: Training with Heterogeneous Samples

Most of the Non-IID Federated Learning research is devoted to studying the relationship between some notion on statistical heterogeneity and its effects on convergence rates and generalization error. Here, we are asking another question related to Machine learning : To what extent does the dissimilarity of learned models themselves represent the statistical distance between the datasets that they were trained on?

As previously stated, the main concern in the canonical Federated Learning problem is the generalization performance of the global model on various client domains. In order to look at statistical heterogeneity from a different angle, we decided to turn this problem over and examine the dissimilarity of models that have the exact same architecture but were trained with different datasets at varying degrees of distributional shift. In the case of parametric models such as Neural Networks or Logistic Regression, we could evaluate model similarity through methods such as canonical correlation analysis (CCA) or simple hypothesis testing, but these methods usually have high evaluation complexity and require a parametric representation. Instead, we will use models' generalization errors on the same previously unseen test set as an indicator of divergence from uniformity.

4.6.1 Method

- We have used MNIST and CIFAR10 datasets, both of which contain images from 10 distinct classes and are often used for benchmarking image classification models. Both sets were split into 10 subsets using the (Dirichlet) distribution-based

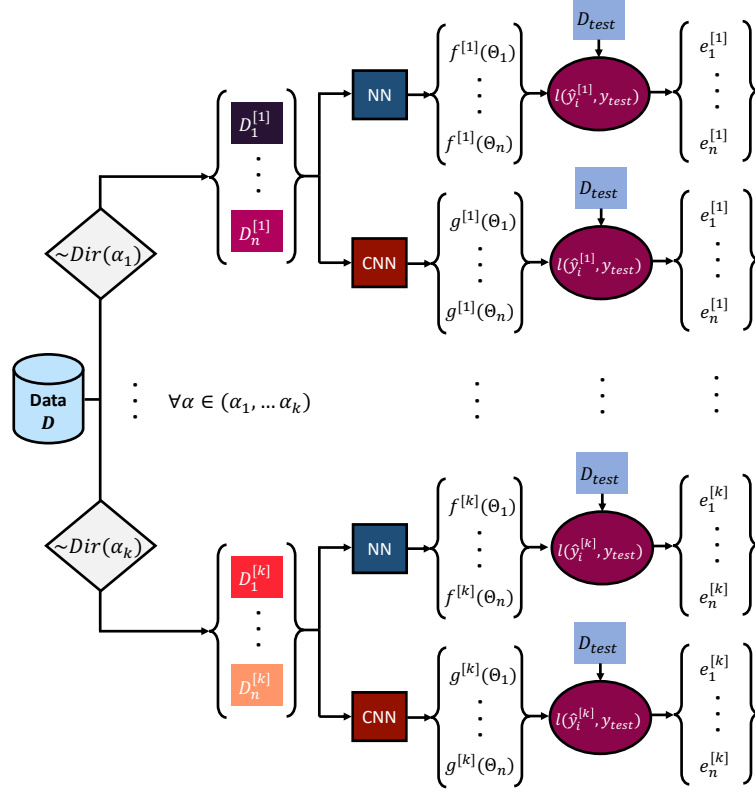


Figure 4.10: Flow chart of training method of NN and CNN models on Dirichlet partitioned data with varying α values.

partitioning approach at varying degrees of heterogeneity $\alpha = (\alpha_1, \dots, \alpha_K)$:

$$P_{Dir}(D; \alpha_i) = \{D_1^{[K]}, \dots, D_{10}^{[K]}\}; \bigcup_{n=1}^{10} D_n^{[K]} = D$$

- Based on our similarity measurements from the previous section, we have chosen $\alpha \in (0, 20]$ as the concentration parameter range since this is the region where most of the change in statistical heterogeneity occurs, with the edge cases $\alpha = 0.1$ and $\alpha = 20$ being close enough to perfect hetero- and homogeneity. Instead of a homogeneous step size, we have opted to choose the individual alpha values

logarithmically from (0,20] in order to achieve reasonable computation time.

$$\alpha \approx (0.1, 0.13, 0.17, 0.23, \dots, 5.13, 20.0); K = 20$$

- Then, we fed these subsets individually into two separate machine learning pipelines, one with a Neural Network (NN) and the other with a Convolutional Neural Network (CNN), to obtain two sets of models for each subset and α -value.

$$train_{NN}(D_n^{[k]}) = f^{[k]}(\theta_n), \text{ where } n \in 10 \text{ and } k \in [200]$$

$$train_{CNN}(D_n^{[k]}) = g^{[k]}(\theta_n), \text{ where } n \in 10 \text{ and } k \in [200]$$

- We have evaluated both set of models by measuring their prediction accuracy on the respective test sets and computed the average loss for each α_k to obtain value pairs (α_k, μ_k) .

$$test(f^k(\theta_n), D_{test}) = e_n^{[k]}, test(g^{[k]}(\theta_n), D_{test}) = e_n^{[k]}$$

$$\mu_k = \frac{1}{n} * \sum_{j=1}^k e_j^{[k]}, \text{ where } n = 10, \text{ and } k \in [K] := [200]$$

- We then plotted these value pairs (α_k, μ_k) to identify an overall behavior.

4.6.2 Model Architecture

Neural Network

We used a simple model architecture with only three fully connected layers:

1. The first layer serializes the spatial image representations (28x28x1 tensors for MNIST and 32x32x3 tensors for CIFAR10) into one-dimensional vectors.
2. Serialized data produced by the first layer flows into a fully connected hidden layer with 256 units and relu activation function.
3. The second and final fully-connected hidden layer has 256 units and relu activation function ($f(x) = \max(0, x)$).
4. A dropout layer was included after the activation of the second hidden layer for regularization purposes.

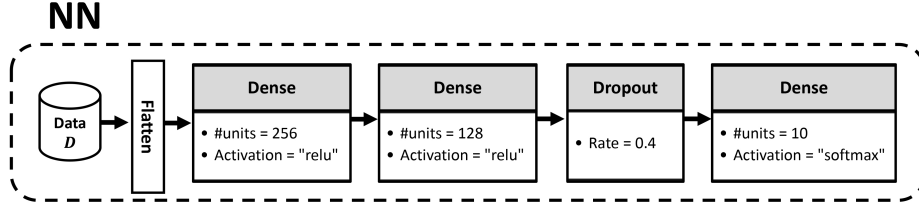


Figure 4.11: Schematic portrayal of the layer architecture of the used neural network.

5. Finally, we have the output layer modeled as a fully-connected layer with 10 units and a softmax activation function $(f(z))_i = \frac{e^{z_i}}{\sum_{j=1}^{|C|} e^{z_j}}$.

We constructed, trained (over 10 epochs at a learning rate of $\eta=0.03$) and evaluated our models using the keras library. The choice of this specific architecture and hyperparameter values was based on available resources on classification with MNIST [26] in order to have a point of reference.

Convolutional Neural Network

In order to enhance the robustness of our findings, we augmented our experiment with a CNN modeling approach. The architecture comprises of a simple alternating pattern of convolutional and pooling layers. The convolutional layers are responsible for extracting patterns from spatial data, whereas pooling layers aggregate information in neighboring features such that most of the relevant information is retained while still achieving dimensionality reduction.

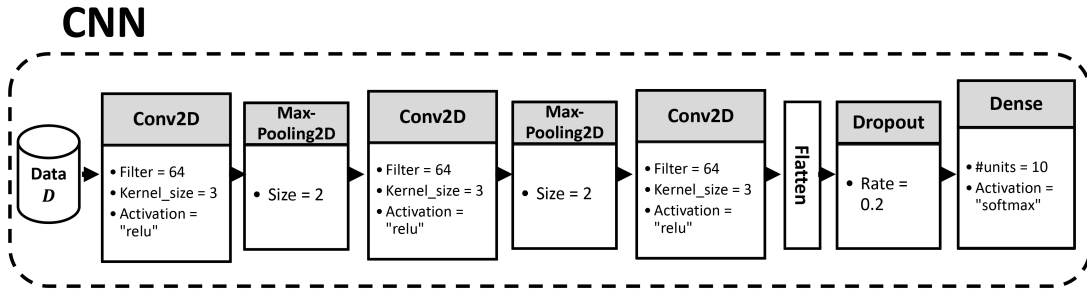


Figure 4.12: Schematic portrayal of the layer architecture of the used convolutional neural network.

4.6.3 Training results on MNIST data

The idea that motivated this experiment was that machine learning models could be treated as compressed representations of the data they were trained on, and compared on the basis of their generalization accuracies on the same domain. A significant correlation between measured statistical distances and error rates would confirm the relationship between model dissimilarity and data dissimilarity.

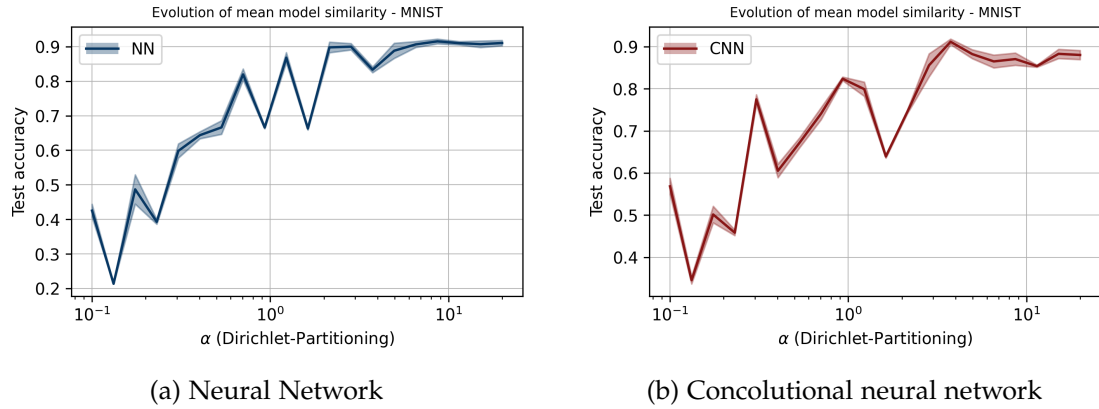


Figure 4.13: Mean successful client model prediction accuracy on Dirichlet partitioned MNIST data as function of the α parameter.

The results we obtained with MNIST do seem to confirm this link: Prediction accuracy in case of extreme domain shift ($\alpha \rightarrow 0$) drops as low as 0.2, which means the models predict on average only two out of ten instances. Approximately at $\alpha = 5$, mean model accuracy beings to converge towards 0.9, which is acceptably close to the maximum test accuracy of 0.98 [27] achieved with this model architecture in fifty (instead of ten) epochs with homogeneous test- and training sets.

In order to establish the link between test accuracy and statistical dissimilarity in numerical terms, we plotted test accuracy results with respect to the four different distance metrics we have examined in the previous section, as can be seen in 4.14.

These numeric results regarding the relationship between model accuracy and distance metrics can be put to use as heuristics in imposing heterogeneity limits on multi-view environments where distributional information is available, or to establish referential equivalence between these distance metrics. Though it must be mentioned that our measurements relating to the distance metrics themselves did exhibit some variability across datasets, and the exact numerical values can only be applicable in similar contexts with similar data distributions.

This set of results also revealed some unexpected behavior that we suspect to be of

algorithmic origin: Whereas the Neural Network pipeline showed a more consistent and stable overall trend, the CNN pipeline exhibited strong fluctuations. The results seen in 4.13b were obtained through subsequent filtering of outliers (test accuracy results under 0.12 were removed), which we have designated as failed experiments. The results for the MNIST&CNN pipeline, as well as an overview of outlier frequencies in different α -regions can be found in figure 4.15.

As stated, the irregularities found in the CNN pipeline are likely due to an algorithmic oversight on our behalf, and we don't attribute meaningful information to them as they appear to be sporadic and inconsistent. Unfortunately, due to time limitations, we were unable to investigate the exact cause of these irregularities. Thus, we have opted to exclude the CNN method from the subsequent stage of our experiment.

4.6.4 Training results on CIFAR10 data

Although the results obtained from the CIFAR10&NN pipeline do exhibit the same overall trend, albeit to a very limited extent: the increase in accuracy is very small, and never goes beyond "slightly better than random guessing". This might be due to the fact that image data in CIFAR10 has more complexity compared to MNIST (real-life colored images of various objects as opposed to grayscale images of handwritten digits), and the representational power of our neural network architecture wasn't sufficient enough to capture it. A certain contributing factor was that we had to restrict the number of training epochs due to time constraints, which effectively limited the achievable level of test accuracy. As a result, our final evaluation will solely rely on results from the MNIST&NN pipeline.

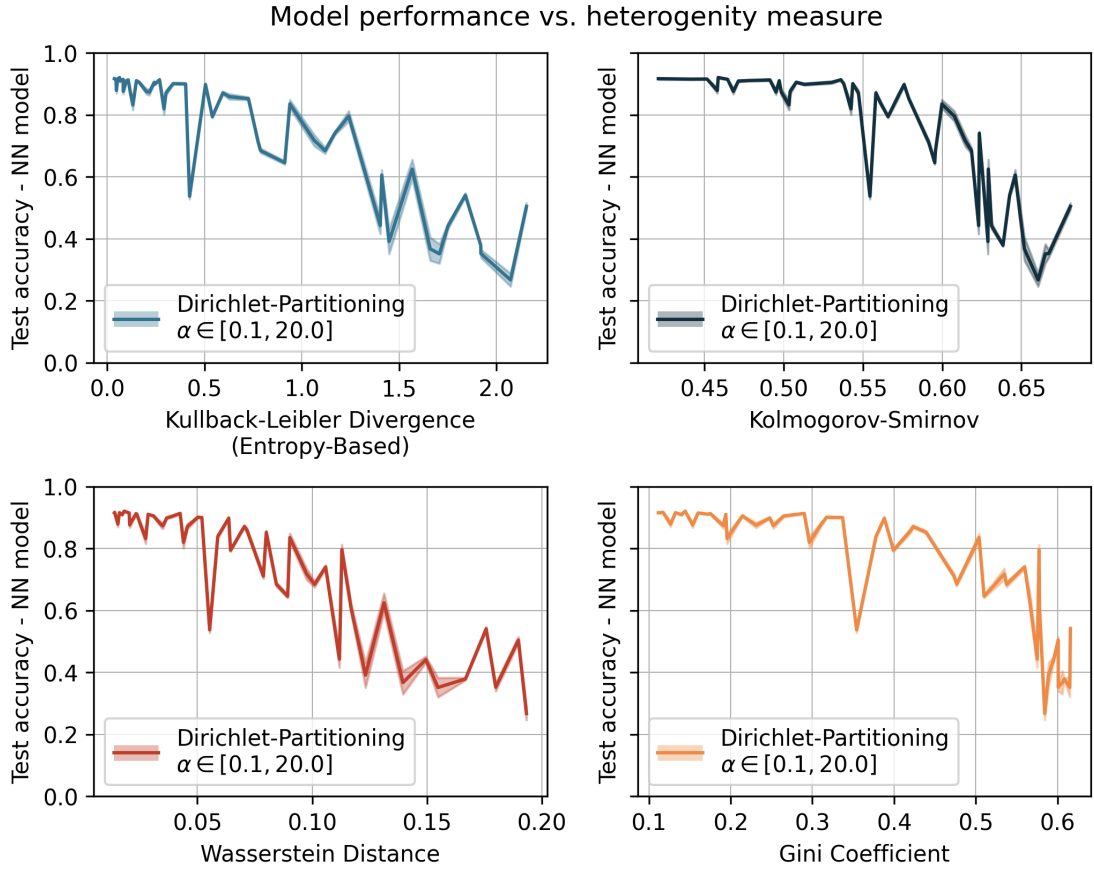


Figure 4.14: The prediction accuracy of neural networks trained on Dirichlet partitioned ($\alpha \in [0.1, 20]$) MNIST data as a function of the mean distance metrics obtained for the same partitions.

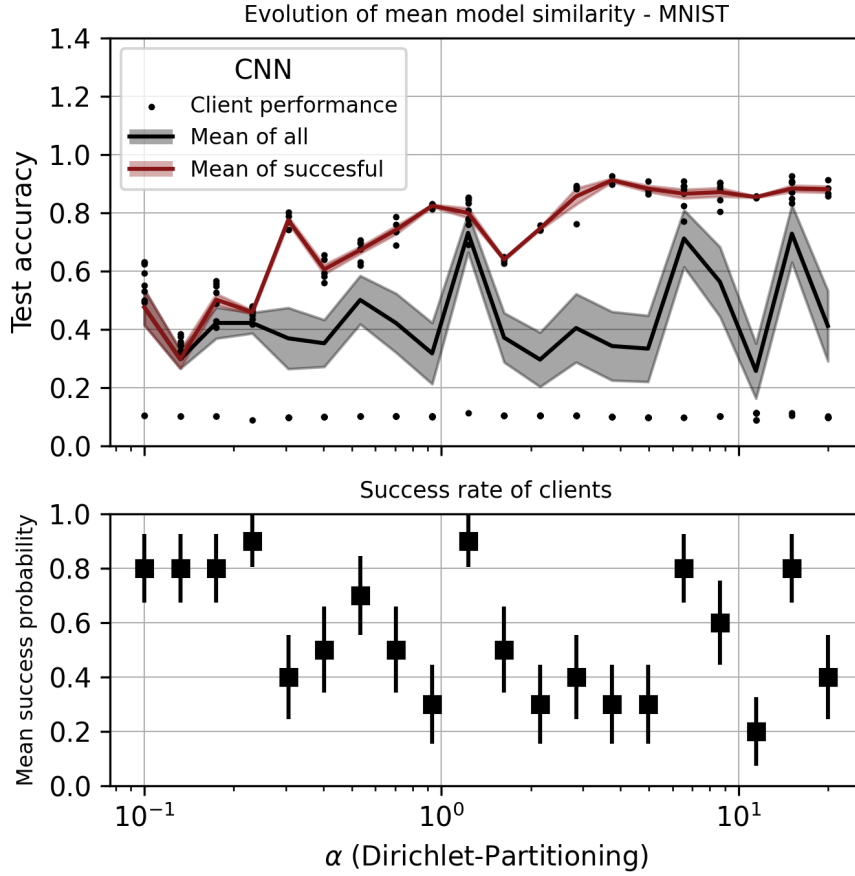


Figure 4.15: (top) Client performances in convolutional neural network prediction accuracy. Mean is given for all clients and for only successful clients, defined as test accuracy greater than 0.2. (bottom) Observed client success probabilities for given α parameter.

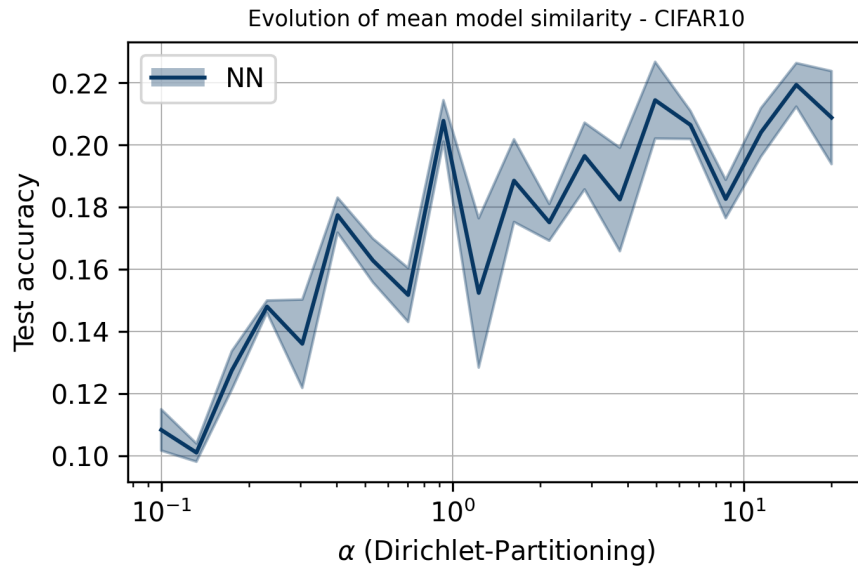


Figure 4.16: Mean successful client neural network model prediction accuracy on Dirichlet partitioned CIFAR10 data as function of the α parameter.

5 Evaluation and discussion

The availability of reliable, reproducible methods for simulating heterogeneous environments is crucial for the study of Non-IID’ness in Federated Learning. In this study, we have confirmed the broader-scale effectiveness of distribution- and quantity-based approaches. On the other hand, we have demonstrated that the quantity-based approach is more sensitive to differences in data properties and can exhibit undefined local patterns. This finding highlights the importance of paying closer attention to how different simulation strategies work in the context of the given domain.

Furthermore, we have shown that different distance metrics show some differences in behavior, which can be an important insight especially if the given domain is operated on a finer scale. Using a combination of distance metrics and partitioning strategies that we identified as more robust and consistent; we measured the link between model and data dissimilarity through a non-parametric approach. The results obtained through these measurements confirmed this link and helped us obtain reference values for half-bounded dissimilarity metrics such as Kullback-Leibler divergence and Wasserstein distance, which can aid the interpretation of their values in similar contexts.

6 Outlook

Our research primarily focused on the general notion of data similarity. However, it is important to note that there is currently a gap in the availability of a reliable similarity metric specifically tailored for Federated Learning environments. The construction of such a metric is impeded by data sharing constraints inherent within the federated setting – only gradient information is shared. During our investigation, we encountered studies examining the question whether the similarity of neural networks can be learned through a Canonical Correlation Analysis (CCA) of their representations [28]. Although this problem statement isn't directly related to ours, the kernel methods discussed there can be used in comparing gradients, as these can be regarded as “representation deltas”. Since our study did not directly delve into representational similarity, we refrain from making a definitive statement regarding the feasibility of employing such an approach in the context of our research, but we strongly encourage other researchers interested in this topic to further explore its potential applications.

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